# Two Hidden Layers Suffice: The Caterpillar Net

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**Abstract**

Every continuous piecewise linear (CPWL) function is a finite signed sum of maxima of affine functions, making maximum representation a central problem in the study of the expressiveness of ReLU networks. Earlier work disproved the conjectured base-two depth bound, leaving open whether a constant number of hidden layers suffices to represent the maximum. Independently and concurrently, Brunck, Hertrich, Rueß, Stade, and Yehudayoff established that two hidden layers suffice for all CPWL functions. We give the *Caterpillar Net*, a non-recursive explicit construction representing the maximum of $N$ real inputs exactly with two hidden layers of widths $\binom{N}{2} + N + 1$ and $2^{O(N \log N)}$, rational weights, and zero biases. Every expansion of the maximum in its finite vocabulary compiles with the same first hidden layer. Sharing its first hidden layer across a Wang–Sun representation gives two-hidden-layer networks for all CPWL functions. [1]

## 1 Introduction

In this paper we investigate the representation capabilities of ReLU networks: what functions are representable by such networks of different depths and widths? ReLU networks represent precisely the continuous piecewise-linear (CPWL) functions with finitely many pieces [ABMM18]. A sharp question is therefore: given a CPWL function, what is the smallest depth, or the smallest width at a prescribed depth, needed to represent it exactly [HBD+23, Saf26]? The maximum is a universal building block for such functions: every CPWL function is a finite signed sum of maxima of affine functions [WS05]. Write

$$\max_N(x) := \max_{0 \le i < N} x_i. \tag{1}$$

**Theorem** (Wang–Sun, [WS05], `Lean_1`)**.** *Let $d \ge 1$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous and affine on each member of a finite polyhedral cover. There are $R \in \mathbb{N}$, affine maps and signs*

$$A_t : \mathbb{R}^d \longrightarrow \mathbb{R}^{d+1}, \qquad \varepsilon_t \in \{-1, 1\} \qquad (1 \le t \le R) \tag{2}$$

*such that*

$$f(x) = \sum_{t=1}^{R} \varepsilon_t \max_{d+1}(A_t(x)) \qquad \textit{for every } x \in \mathbb{R}^d. \tag{3}$$

[1] Our results were obtained over months of proof search in Lean using Claude Code and Codex; the AI methodology statement describes their roles. The Lean code is available in the accompanying repository https://github.com/JayPiZimmermann/Max_Caterpillar. Links to the Lean formalization appear in theorem titles and equation numbers; equation numbers are clickable.

Thus a two-hidden-layer representation of $\max_{d+1}$ gives one for every CPWL function on $\mathbb{R}^d$: precompose with the affine maps and add the networks in parallel. Section 8 reuses their first hidden layer; the representation theorems and the resulting networks are formalized.

**Our contribution.** The *Caterpillar Net* gives a nonrecursive two-hidden-layer representation of the maximum with explicit bounds on both hidden-layer widths. The construction uses a finite dictionary of support functions of caterpillar polytopes and their faces. The maximum is the support function of a simplex. We first express this support function as an alternating sum of caterpillar support functions, up to a linear combination of support functions of lower-dimensional polytopes. We then represent the remaining error using faces of the same caterpillar polytopes and elementary ReLU features. Normalizing and merging these descriptions yields a width bound for the complete network, including all correction terms.

Write $\sigma(t) := \max(0, t)$ applied coordinatewise to vectors. A network has an input layer of $N$ coordinates, first and second hidden layers of widths $h_1$ and $h_2$, respectively, and a scalar output layer. With real weight matrices $W_1, W_2$, output vector $w_3$, and biases $b_1, b_2, b_3$ of the corresponding dimensions, it evaluates as

$$x \longmapsto w_3^\top \sigma\big(W_2\sigma(W_1x + b_1) + b_2\big) + b_3. \tag{4}$$

There are no skip connections; equality is required on all of $\mathbb{R}^N$. A network has *rational weights* if every entry of $W_1$, $W_2$, and $w_3$ is rational, and *zero biases* if $b_1$, $b_2$, and $b_3$ vanish.

**Theorem 1** (Caterpillar Net, `Lean_2`, Proof)**.** *For every $N \geq 3$, there is a two-hidden-layer ReLU network with rational weights and zero biases computing $\max_N$ with widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = 3^{N-2}C_N - N(N-1) + 2, \tag{5}$$

*where*

$$C_N := \sum_{m=2}^{N-1} \binom{N}{m} \frac{m!}{2}(N-m)m^{N-m-1}. \tag{6}$$

*Thus $h_1 = O(N^2)$ and $h_2 = 2^{O(N \log N)}$ as $N \to \infty$.*

**Further results.** The network is compiled from an expansion of $\max_N$ in a finite vocabulary of caterpillar face functions. Theorem 6 separates the network from the coefficients: any such expansion with $|K|$ retained units compiles with the same first hidden layer and $|K| + 2$ second hidden layer neurons. The coefficients are the coordinates of $\max_N$ in one basis of the vocabulary; the same proof shows that they are rational, without any integral (Theorem 15). Section 7 bounds the second width by $q_N + 2$, where $q_N$ is the dimension of the vocabulary modulo its auxiliary functions; this bound proves Theorem 1. Conversely, Section 4 proves that every expansion of $\max_N$ in caterpillar and auxiliary functions has at least $(7/6)^N/(N+1)$ caterpillar functions, and that $q_N \geq (N-2)!$. Section 8 applies the construction to sums of maxima.

## 2 Related work

The conjecture that $\max\{0, x_1, \ldots, x_{2^k}\}$ requires more than $k$ hidden layers [HBD+23] was disproved for $k = 2$ by the maximum of five inputs in [BBH+26]. Exact computer-assisted searches subsequently obtained two-hidden-layer representations up to $N = 12$ in [RAB+26] and, independently, up to $N = 8$ in [WB26]. The improved general depth bounds remained logarithmic, leaving constant depth open.

Setting all biases to zero does not change a network that computes a positively homogeneous function such as $\max_N$ [HBD+23, Proposition 2.3], [HHL23, Lemma 7], so biases give no additional power for representing the maximum. Depth lower bounds hold for integer weights [HHL23] and rational weights whose denominators are restricted to powers of a fixed integer [AHM25]. Our weights are rational, but their denominators are not restricted in this way, so neither bound applies. Other results concern braid-fan-compatible networks [GHL25] and first hidden layer width bounds [Saf26, Lu26].

**Theorem 2** (Brunck, Hertrich, Rueß, Stade, and Yehudayoff, [BHR+26])**.** *Every continuous piecewise affine function $f : \mathbb{R}^d \to \mathbb{R}$ with finitely many affine pieces can be represented exactly by a ReLU network with two hidden layers.*

Their first proof proceeds by induction on dimension, using homological algebra to construct a linear combination of support functions of pyramids over zonotopes that agrees with the simplex support function modulo lower-dimensional terms. Our construction restricts these building blocks to an explicitly specified finite family, the caterpillar polytopes; we deliver a geometric proof of the corresponding identity.

Independently and concurrently, they developed a second proof based on combinatorial cancellation, with building blocks closely related to ours. Appendix D identifies them exactly: for nonempty marked sets their pyramids are translated caterpillar polytopes, and the remaining pyramids are rooted stars, which are not caterpillar functions. We do not claim that their coefficients agree with ours. We establish explicit width bounds for the first and second hidden layers; their paper states no widths.

# 3 The Caterpillar Net

Two identities drive the construction:

$$\max(s, t) = t + \sigma(s - t), \tag{7}$$

$$\sigma(\ell(x)) = \sigma(-\ell(x)) + \ell(x). \tag{8}$$

The first suggests a path with attached leaves. A signed sum of the resulting caterpillar support functions agrees with the maximum modulo lower-dimensional terms; proper faces and linear functions correct the error. The second identity reverses hinges to merge units in Section 6.

Fix $N \geq 3$ and write $x = (x_0, \ldots, x_{N-1}) \in \mathbb{R}^N$.

**Definition 3** (Labelled caterpillar, `Lean_3`)**.** A *caterpillar description* is a permutation–set pair

$$\pi \in \mathrm{Sym}\{0, \ldots, N-1\}, \qquad \varnothing \neq S \subseteq \{1, \ldots, r\}, \qquad r := N - 2. \tag{9}$$

Position $j$ carries label $\pi(j+1)$, and $\pi(0)$ is the *apex*. The *spine positions* are $\{0\} \cup S$. The *cut set* $S$ separates positions $j-1$ and $j$ whenever $j \in S$. The last spine position $q$ and the *parent* $p(j)$, the nearest preceding spine position, are

$$q := \max S, \qquad p(j) := \max\big(\{0\} \cup \{i \in S : i < j\}\big) \qquad (1 \leq j \leq r). \tag{10}$$

The parent–child edges form the *labelled caterpillar*: a path through the spine positions with each remaining label attached as a leaf to its parent (Figure 1). The apex is separate from the tree.

**Definition 4** (Caterpillar function and face functions, `Lean_4`, `Lean_5`)**.** For labels $i, j$, for the tree edges, and for the head, put

$$\ell_{ij}(x) := x_i - x_j, \tag{11}$$

$$L_{\pi,S} := \{\ell_{\pi(p(j)+1),\pi(j+1)} : 1 \le j \le r\}, \tag{12}$$

$$h_{\pi,S} := \ell_{\pi(q+1),\pi(0)}. \tag{13}$$

The *edge functions* $L_{\pi,S}$ subtract each child's coordinate from its parent's, one for each of the $r$ tree edges, and the *head function* $h_{\pi,S}$ compares the last spine vertex with the apex. For a linear function $h$, a finite set $L$ of linear functions, and a *state* $\tau(\ell) \in \{0,\ \ell,\ \sigma \circ \ell\}$ chosen for every $\ell \in L$, define

$$U^{\tau}_{h,L}(x) := \sigma\Big(h(x) + \sum_{\ell \in L} \tau(\ell)(x)\Big). \tag{14}$$

Each edge is omitted, kept linearly, or kept as a hinge. Write $U_{h,L}$ for the all-hinge state, and call $(h, L)$ its *function description*, recording the head and inner hinges. The *caterpillar function* of $(\pi, S)$ and its *face functions* are

$$U_{\pi,S} := U_{h_{\pi,S},L_{\pi,S}}, \qquad U^{\tau}_{\pi,S} := U^{\tau}_{h_{\pi,S},L_{\pi,S}}. \tag{15}$$

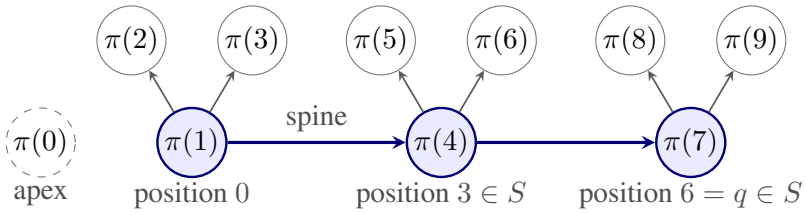



Figure 1: A labelled caterpillar with $N = 10$ and $S = \{3, 6\}$. The spine $\pi(1) \to \pi(4) \to \pi(7)$ has two leaves at each vertex; all arrows point from parent to child. The apex $\pi(0)$ is separate from the tree. The last spine position is $q = 6$, so the head function is $h_{\pi,S}(x) = x_{\pi(7)} - x_{\pi(0)}$.

The term *face function* comes from the following geometry. Identify $a(x) = \langle a, x\rangle$ with its coefficient vector. Then $\ell_{ij} = e_i - e_j$ points from $e_j$ to $e_i$, and $h = h_{\pi,S} = e_{\pi(q+1)} - e_{\pi(0)}$ points from the apex label to the last spine label. Each $\ell \in L = L_{\pi,S}$ points from child to parent. As free vectors, $h$ places a base corner and the $\ell$ generate its edges. The polytope apex is 0, distinct from the labelled point $e_{\pi(0)}$. Define the *support function* and the caterpillar polytope by

$$h_P(x) := \max_{p \in P} \langle p, x\rangle, \tag{16}$$

$$Q_{h,L} := \operatorname{conv}\Big(\{0\} \cup \Big(h + \sum_{\ell \in L} [0, \ell]\Big)\Big), \qquad U_{h,L} = h_{Q_{h,L}}. \tag{17}$$

Indeed, the segment $[0, \ell]$ has support function $\sigma \circ \ell$, Minkowski sums add support functions, and adjoining 0 applies the outer ReLU. The base is the translated zonotope $h + \sum_{\ell \in L}[0, \ell]$. For a state $\tau$ in Equation (14), set $t_\ell = \mathbb{1}_{\{\tau(\ell) = \ell\}}$, where $\mathbb{1}$ denotes the indicator. It is zero for omitted edges and hinges; the latter contribute segments. The corresponding polytope is $Q^{\tau}_{h,L} := \operatorname{conv}\big(\{0\} \cup (h + \sum_{\ell \in L} t_\ell \ell + \sum_{\tau(\ell) = \sigma \circ \ell}[0, \ell])\big)$, and $U^{\tau}_{h,L} = h_{Q^{\tau}_{h,L}}$. Figure 3 illustrates the translated generators for $N = 4$ on the slice $x_0 = 0$, where $a$ becomes $(a_1, a_2, a_3)$ and $e_0$ is represented by 0.

The head connects the apex label to the tree, so the head and $N-2$ edge functions are linearly independent. Thus the base is an $(N-2)$-dimensional zonotope with affine hull avoiding 0, and $Q_{h,L}$ is an $(N-1)$-dimensional pyramid. Each nontrivial face containing 0 cones over a base face. Maximizing on each base segment selects 0, $\ell$, or $[0, \ell]$, with support functions 0, $\ell$, and $\sigma \circ \ell$; the outer ReLU adjoins the apex. These are precisely the three states in Equation (14).

**Definition 5** (Dictionary, `Lean_6`)**.** The *dictionary* $\mathcal{D}$ is the finite set of functions $\mathbb{R}^N \to \mathbb{R}$ consisting of every face function $U^\tau_{\pi,S}$, over all caterpillar descriptions $(\pi, S)$ and all states $\tau$, together with the *auxiliary functions*

$$\begin{aligned} &x \mapsto \sigma(x_i), \quad x \mapsto \sigma(-x_i) \quad (0 \le i < N), \\ &x \mapsto \sigma(x_i - x_j) \qquad\qquad\quad (0 \le i < j < N). \end{aligned} \tag{18}$$

Heads and edge functions are coordinate differences, so face functions and the functions $x \mapsto \sigma(x_i - x_j)$ are translation invariant: with $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^N$,

$$U^\tau_{\pi,S}(x + t\mathbf{1}) = U^\tau_{\pi,S}(x) \qquad (t \in \mathbb{R}). \tag{19}$$

Only $\sigma(\pm x_i)$ change under this translation. Theorem 14 proves that $\mathcal{D}$ spans the maximum. Section 7 chooses rational coefficients from one basis of its span, without integration.

### 3.1 The network

The network computes an expansion of $\max_N$ in face functions and auxiliary functions. To compile face functions, first absorb every linear edge of a face state into the head:

$$U^\tau_{h,L} = U_{h',M}, \qquad h' := h + \sum_{\tau(\ell)=\ell} \ell, \qquad M := \{\ell \in L : \tau(\ell) = \sigma \circ \ell\}. \tag{20}$$

By Equation (8), reversing each remaining $\ell_{ij}$ with $i > j$ gives the same function with the *normalized description*

$$\begin{aligned} M^+ &:= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in M\}, \\ h^+ &:= h' + \sum_{\ell_{ij} \in M,\ i>j} \ell_{ij}. \end{aligned} \tag{21}$$

Geometrically, $[0, \ell] = \ell + [0, -\ell]$: reversing a generator shifts the head by the old generator. Normalization chooses a consistent orientation and a corresponding base corner; it does not rescale vectors or change the polytope (Figure 3). Write $U_k := U_{h,M}$ for a normalized description $k = (h, M)$, and let $\mathcal{K}_0$ be the set of normalized descriptions of all face functions. Let $A_N$ be the span of the auxiliary functions Equation (18). The network has three parts (Figure 2).

- The *first hidden layer* computes the shared features

$$\Phi(x) := \big(\sigma(x_0),\ \sigma(-x_0),\ (\sigma(x_i - x_0))_{i>0},\ (\sigma(x_i - x_j))_{i<j}\big). \tag{22}$$

- The *second hidden layer* computes the retained units $U_k$, $k \in K$, for a set $K \subseteq \mathcal{K}_0$, and the two neurons $\sigma(R)$ and $\sigma(-R)$ for an auxiliary remainder $R \in A_N$.
- The *output layer* returns $\sum_{k \in K} d_k U_k + \sigma(R) - \sigma(-R)$.

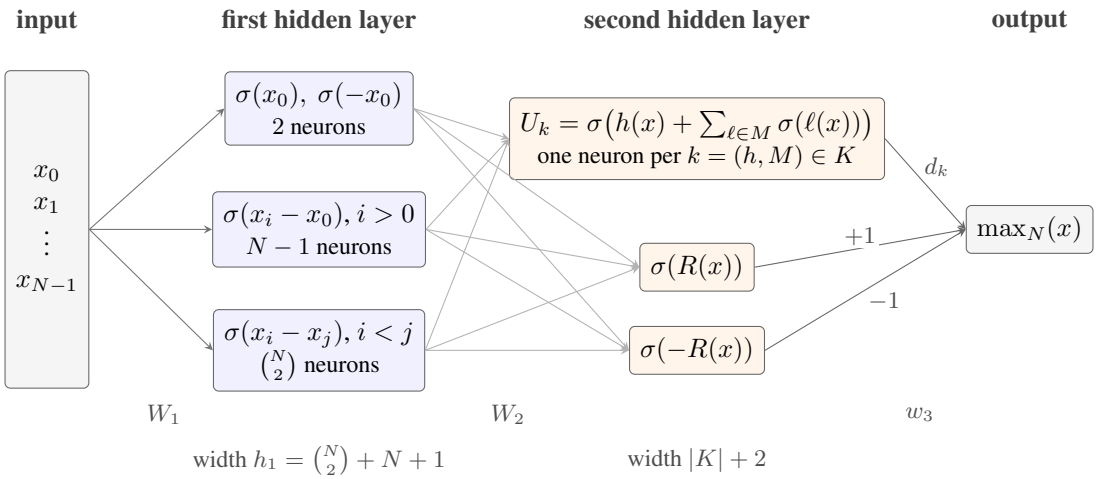



Figure 2: The Caterpillar Net. The first hidden layer is fixed. The second hidden layer has one neuron for every retained normalized description $k = (h, M)$ and two neurons for the auxiliary remainder $R$. All biases are zero; the matrices are given in Equation (29).

**Theorem 6** (Compilation, `Lean_7`, Proof)**.** *Let $K \subseteq \mathcal{K}_0$, $d \in \mathbb{R}^K$, and $R \in A_N$ satisfy*

$$\max_N = \sum_{k \in K} d_k U_k + R. \tag{23}$$

*Then the network with first hidden layer Equation* (22) *and the matrices in Equation* (29) *computes* $\max_N$*. It has widths $h_1$ and $|K| + 2$ and zero biases. Its weights are rational whenever the coefficients $d_k$ and the auxiliary coefficients of $R$ are rational.*

**Proof of Theorem 6.** The first hidden layer outputs $\Phi$ from Equation (22). Thus $W_1$ has rows $e_0^\top$ and $-e_0^\top$, then $e_i^\top - e_0^\top$ for $i > 0$, then $e_i^\top - e_j^\top$ for $i < j$, using a fixed ordering within each block. Its row count is

$$\binom{N}{2} + (N-1) + 2 = \binom{N}{2} + N + 1 = h_1. \tag{24}$$

By Equation (8), these features recover all coordinates,

$$\begin{aligned} x_0 &= \sigma(x_0) - \sigma(-x_0), \\ x_i &= x_0 + \sigma(x_i - x_0) - \sigma(x_0 - x_i) \qquad (i > 0). \end{aligned} \tag{25}$$

For each $k = (h, M) \in K$, substitute Equation (25) into $h(x) + \sum_{\ell \in M} \sigma(\ell(x))$ and collect the coefficients of $\Phi(x)$ into a row vector $w_k^\top$. Every hinge in $M$ already has the orientation $i < j$, so its coefficient is added directly to the corresponding feature.

To specify the remaining row, collect the auxiliary coefficients of $R$ as

$$R(x) = \sum_i p_i \sigma(x_i) + \sum_i n_i \sigma(-x_i) + \sum_{i<j} s_{ij} \sigma(x_i - x_j). \tag{26}$$

Each retained unit is translation invariant by Equation (19), and $\max_N(x + t\mathbf{1}) = \max_N(x) + t$, so Equation (23) gives

$$R(x + t\mathbf{1}) = R(x) + t, \qquad \sum_i p_i = 1. \tag{27}$$

The coefficient sum follows by setting $x = 0$ and $t = 1$. Now choose $t$ large enough that every $x_i + t$ is nonnegative. Substituting the auxiliary expansion into Equation (27) cancels $t$ and gives

$$R(x) = \sum_i p_i x_i + \sum_{i<j} s_{ij}\sigma(x_i - x_j). \tag{28}$$

Substitute Equation (25) into this expression and collect the coefficients of $\Phi(x)$ into $w_R$. Thus both $w_k$ and $w_R$ are specified by finite sums of the previously defined coefficients. The entries of $W_1$ and of the rows $w_k$ are integers, because heads and hinges are integer combinations of coordinates. The row $w_R$ is a rational combination of the $p_i$ and $s_{ij}$, and $w_3$ consists of the $d_k$ and $\pm 1$.

Order $K$ and set the second hidden layer matrix and output layer vector to

$$W_2 = \begin{pmatrix} (w_k^\top)_{k\in K} \\ w_R^\top \\ -w_R^\top \end{pmatrix}, \qquad w_3 = \begin{pmatrix} (d_k)_{k\in K} \\ 1 \\ -1 \end{pmatrix}, \qquad b_1 = b_2 = b_3 = 0. \tag{29}$$

The second hidden layer outputs $U_k(x)$ and $\sigma(\pm R(x))$. By Equation (8) and Equation (23),

$$w_3^\top \sigma(W_2\Phi(x)) = \sum_{k\in K} d_k U_k(x) + \sigma(R(x)) - \sigma(-R(x)) = \max_N(x). \tag{30}$$

□

Theorem 15 gives such an expansion with rational coefficients; Section 6 counts the normalized descriptions that can occur.

# 4 Lower bounds

The second width of the network of Theorem 6 is the number of caterpillar functions in an expansion of $\max_N$, plus two. The lower bounds below share a local test: at every tied maximum, some unit must retain the top-dimensional simplex class. Coefficient balance bounds how often the restricted orientations can do this; the rank of the retained comparisons gives a second bound for arbitrary orientations and weights. The same test lets us combine the two families. We also bound the dimension of the vocabulary.

**Definition 7** (Caterpillar expansion, `Lean_8`)**.** A finite set $K \subseteq \mathcal{K}_0$ is a *caterpillar expansion* of $\max_N$ if there are real numbers $d_k$, $k \in K$, and a function $R \in A_N$ with $\max_N = \sum_{k\in K} d_k U_k + R$ on $\mathbb{R}^N$. Its *size* is $|K|$.

**Theorem 8** (Exponential count, `Lean_9`, `Lean_10`, `Lean_11`, Proof)**.** *Let $N \geq 3$ and let $K$ be a caterpillar expansion of $\max_N$. Then*

$$\binom{N}{m} \leq 6^m\,|K| \qquad (3 \leq m \leq N), \tag{31}$$

$$7^N \leq (N+1)\,6^N\,|K|, \tag{32}$$

$$N(N-1) \leq 6\,|K|. \tag{33}$$

*In particular $|K| \geq (7/6)^N/(N+1)$, and every network of the form of Theorem 6 for $\max_N$ has second width at least $(7/6)^N/(N+1) + 2$.*

The bound Equation (32) follows from Equation (31) by summing over all $m$ with the binomial theorem; it exceeds the quadratic bound Equation (33) only for $N$ of order 70. At an $m$-coordinate top tie, the common local test in Lemma 16 kills shallow functions and leaves a nonzero simplex class. Some unit must therefore survive. For a caterpillar function, coefficient balance and a partition of its reduced forest into small stars bound the fraction of surviving localities by $6^m/\binom{N}{m}$. The same argument applies to a larger vocabulary. For an integer head $h$ and a set $M$ of oriented edges of a forest, write $U(x) = \sigma(\sum_i h_i x_i + \sum_{(i,j)\in M} \sigma(x_i - x_j))$ and $\kappa_i = h_i + |\{j : (i,j) \in M\}|$. Delete, simultaneously, all degree-one vertices with $\kappa_i = 0$ from the forest; call the result its *reduced forest*.

**Theorem 9** (Forest expansions, `Lean_12`, Proof)**.** *Let $\Delta \geq 2$ be an integer. Suppose $\max_N = \sum_{k\in K} d_k U_k + R$, where $R \in A_N$ and every $U_k$ has an integer head and a forest of functions $\sigma(x_i - x_j)$ whose reduced forest has maximum degree at most $\Delta$. Put $b = 2\Delta + 2$. Then*

$$\binom{N}{m} \leq b^m |K| \qquad (3 \leq m \leq N). \tag{34}$$

*Consequently $|K| \geq (1 + 1/b)^N/(N+1)$ for $N \geq 3b + 2$. For $\Delta = 2$, the latter bound holds for every $N \geq 3$.*

Caterpillar functions have reduced degree at most 2: their retained leaf edges have $\kappa_i = 0$ at the leaf, and their remaining edges lie on a path. The rooted stars $\sigma(x_c - x_a + \sum_{l\neq a,c} \sigma(x_c - x_l))$ and all their face functions have reduced degree 0. Thus the bound $(7/6)^N/(N+1)$ counts all functions in any mixture of these families, including the building blocks of the direct combinatorial construction of Brunck et al. [BHR+26]; see Appendix D. More generally, Theorem 9 covers every tree of bounded degree, with arbitrary edge orientations, and permits unboundedly many deleted leaves at each vertex.

**Theorem 10** (Paths and cycles with arbitrary weights, Proof)**.** *Suppose $N \geq 3$ and $\max_N = R + \sum_{k\in K} d_k U_k$, where $R$ is a linear function plus a finite linear combination of ReLUs of linear forms, and*

$$U_k(x) = \sigma\left(h_k(x) + \sum_{(i,j)\in M_k} c_{k,ij}\sigma(x_i - x_j)\right). \tag{35}$$

*The heads $h_k$ are arbitrary linear functions, all coefficients are arbitrary real numbers, and each $M_k$ orients a simple graph of maximum degree at most two. Then*

$$\binom{N}{m} \leq \left(N + \frac{m-1}{2}N^2\right)|K| \qquad (3 \leq m \leq N). \tag{36}$$

*Taking $m = \lfloor N/2 \rfloor$ gives $|K| = \Omega(2^N/N^{7/2})$ as $N \to \infty$. The conclusion remains valid when biases are allowed.*

The comparison graphs in Theorem 10 may be disconnected unions of paths and cycles. The same locality argument extends to *pseudoforests*: graphs with at most one cycle in each connected component.

**Theorem 11** (Weighted comparison graphs, Proof)**.** *Let $N \geq 3$. Use the representation and coefficient assumptions of Theorem 10, with comparison pseudoforests $G_j$. Write $\nu(G_j)$ for their maximum matching sizes and $z(G_j)$ for their numbers of isolated vertices. If $\nu(G_j) + z(G_j) \geq k$ for a nonnegative integer $k$ and every $j$, put $b_N = 1 + N + \binom{N}{2}$ and $c_k = 1 + k/3 + \binom{k}{2}/9$. Then*

$$|K| \geq \frac{1 - b_N 2^{-N}}{c_k}\left(\frac{4}{3}\right)^k. \tag{37}$$

*For arbitrary comparison graphs of maximum degree at most $\Delta \geq 3$, put $y_\Delta = (\Delta-2)^{\Delta-2}/(\Delta-1)^{\Delta-1}$. Then*

$$|K| = \Omega_\Delta\big((1+y_\Delta)^N\big). \tag{38}$$

*These bounds also hold for the number of distinct comparison graphs used.*

Degree in these two theorems refers to the full graph, before any leaf deletion. Every fixed $\Delta$ therefore gives an exponential lower bound, with arbitrary orientations, heads, weights, and biases. The proofs use the rank of the comparisons at a tied maximum and elementary counting; they are not yet formalized in Lean. A matching of linear size in each pseudoforest also gives an exponential bound without restricting the maximum degree. Trees of unbounded degree and sublinear matching number remain outside this exponential regime.

Additional cycles are allowed. Put $b(G) = \sum_C \max\{0, |E(C)| - |V(C)|\}$ for a comparison graph $G$, summing over its connected components. If every graph has $\nu(G) + z(G) \geq k$ and $b(G) \leq b$, then

$$|K| \geq \frac{1 - b_N 2^{-N}}{\Pr\big(\mathrm{Bin}(k, 1/4) \leq b+2\big)}. \tag{39}$$

Thus a matching of linear size and $b = o(N)$ still give an exponential bound, without any degree restriction. More generally it suffices that $k = \Omega(N)$ and $\limsup b/k < 1/4$. The proof is in Appendix C.1.

**Trees with few out-leaves (`Lean_13`).** There is also an exponential bound without restricting internal branching. In the normalized description of a tree face function, call a vertex an *out-leaf* if it has degree one in the retained forest and $\kappa_i = 1$. If every vertex of every tree function is adjacent to at most $s \geq 1$ out-leaves, then any expansion by these functions together with caterpillar functions satisfies $|K| \geq (1 + 1/(2s+4))^N/(N+1)$ for $N \geq 3$, where $K$ counts both families. Thus the relevant restriction concerns the oriented edges and head of each function, rather than just the shape of its original tree.

For example, take *any* tree $T$ on the labels other than an apex $a$, choose vertices $s, r$ of $T$, orient every edge away from $s$, and retain all the functions $\sigma(x_p - x_q)$ on those oriented edges. The resulting function is $U_{a,s,r,T}(x) = \sigma(x_r - x_a + \sum_{p\to q \text{ in } T} \sigma(x_p - x_q))$. At a tree vertex $v$, normalization gives $\kappa_v = \deg_T(v) - 1 + \mathbf{1}_{v=s} + \mathbf{1}_{v=r}$. Only $s$ and $r$ can therefore be out-leaves. A degree-two covering forest improves the bound in this case to $(7/6)^N/(N+1)$, even when these functions are added to the caterpillar and rooted-star face dictionaries; the argument is given at the end of Appendix C.1. This includes every underlying tree shape. It does not include arbitrary face functions of those added trees: replacing edges by linear terms can create many out-leaves at one vertex.

**Large matchings with arbitrary heads and weights.** The matching argument also works in mixtures containing the original constructions. Let $N \geq 12$ and $0 \leq 2k \leq N$. Start with any mixture of the reduced-degree-two functions in Theorem 9 and the source-oriented functions just described; this includes both the caterpillar and rooted-star face dictionaries. Adjoin functions of the form Equation (35) whose comparison pseudoforest on the $N$ labels has a matching of size at least $k$. These added functions may have arbitrary real heads and coefficients. Every expansion of $\max_N$ in this enlarged dictionary, with an auxiliary remainder, uses a number $Z$ of functions satisfying

$$Z \geq \frac{(16/13)^k}{(N+1)d_k}, \qquad d_k = 1 + \frac{3k}{13} + \frac{9}{169}\binom{k}{2}. \tag{40}$$

The proof at the end of Appendix C.1 is mathematical; this extension is not yet formalized in Lean.

In particular, every orientation and face state of a canonical tree with matching number at least $k$ is covered. Omitted edges may be retained with coefficient zero and linear edge terms absorbed into the head, so

the matching condition concerns the original tree. Taking $k \geq \varepsilon(N-1)$ for fixed $\varepsilon > 0$ gives an exponential bound. This includes all trees with a perfect or near-perfect matching, and every star whose edges are subdivided once. More generally, a tree on $n \geq 2$ vertices with $L$ leaves has a matching of size at least $(n - L + 1)/2$, so a fixed positive fraction of nonleaf vertices suffices. The enlarged dictionary still spans the maximum because it contains the original caterpillar dictionary.

Arbitrary weighted comparison graphs of bounded degree can also be added to both original dictionaries, with no restriction on their number of cycles. For maximum degree at most three, the enlarged dictionary requires $\Omega((7/6)^N/\sqrt{N})$ functions. For any fixed $\Delta \geq 4$, the bound is $\Omega_\Delta((1 + y_\Delta)^N)$. These mixture bounds follow from the same connected-set count in Appendix C.1.

**Corollary 12** (All tree orientations, `Lean_14`)**.** *For $N \geq 3$, every expansion of $\max_N$ by canonical tree face functions and a remainder in $A_N$ has at least $N(N-1)/6$ terms. Here the head is $x_r - x_a$ for distinct root and apex labels; the tree on all labels other than $a$ is arbitrary, and each oriented edge contributes zero, its linear form, or the ReLU of that form. All orientations and face states are allowed, with arbitrary signed output coefficients.*

**Stars with arbitrary orientations.** All star face functions can also be adjoined, with arbitrary head roots, edge orientations, and face states. The caterpillar dictionary together with this entire star family requires $\Omega((7/6)^N/\sqrt{N})$ nonstar terms. The bounded-degree mixture bounds just stated remain valid after adjoining this star family as well, counting only terms outside it. If the matching class above is included, Equation (40) holds for $N \geq 20$, again counting only terms outside the added star family. This follows by propagating an exactly checked five-label separator through coordinate deletions; the argument is given in Appendix C.1. It is not yet a Lean result. Nonstar trees with sublinear matching number remain unresolved in general, so the unrestricted all-tree exponential bound is still open.

**Theorem 13** (Vocabulary dimension, `Lean_15`, Proof)**.** *For every $N \geq 3$, the span $V_N := \operatorname{span} \mathcal{D}$ of the dictionary satisfies $q_N := \dim(V_N/A_N) \geq (N-2)!$.*

Thus Theorem 15 cannot give a second width below $(N-2)! + 2$; together with Theorem 1, $\log q_N = \Theta(N \log N)$. These bounds concern the stated dictionaries; they do not give a width lower bound for general two-hidden-layer networks.

# 5 Why the dictionary spans the maximum

The maximum is the support function of a simplex. To work in its affine dimension, set

$$\Delta := \operatorname{conv}\{0, e_1, \ldots, e_{N-1}\} \subseteq H_0 := \{x \in \mathbb{R}^N : x_0 = 0\}. \tag{41}$$

For every $x \in \mathbb{R}^N$,

$$\max_N(x) = x_0 + h_\Delta(x - x_0\mathbf{1}). \tag{42}$$

It suffices to prove that $h_\Delta$ lies in the span of the restrictions of $\mathcal{D}$ to $H_0$: the substitution $x \mapsto x - x_0\mathbf{1}$ fixes every face function, sends coordinate features to $\sigma(\pm(x_i - x_0))$, and preserves the dictionary span by Equation (8).

Henceforth all polytopes lie in $H_0$, and heads and edge functions are identified with their restricted coefficient vectors. Let $\mathcal{L}_k$ be the real span of support functions of polytopes of affine dimension at most $k$; thus $\mathcal{L}_0$ consists of linear functions. The *exposed face* of $P$ in direction $u \in H_0$ is

$$P^u := \{p \in P : \langle p, u\rangle = h_P(u)\}. \tag{43}$$

The opposite face $P^{-u}$ consists of the minimizers. Define

$$D_u h_P(x) := h_{P^u}(x) - h_{P^{-u}}(x) \qquad (x \in H_0). \tag{44}$$

Here $u$ selects the faces, while $x$ is the support-function input; $P^0 = P$ gives $D_0 = 0$. For a simplex facet normal, $D_u$ subtracts the opposite vertex's linear support function from that of the facet.

For a finite linear combination $g$ of support functions, define its one-sided directional derivative at $u$ by

$$\begin{aligned} \partial_u g(x) &:= \lim_{t \downarrow 0} \frac{g(u + tx) - g(u)}{t}, \\ \partial_u h_P = h_{P^u}, &\qquad D_u g = \partial_u g - \partial_{-u} g. \end{aligned} \tag{45}$$

This makes $D_u$ independent of the support-function decomposition. When $\langle u, \cdot \rangle$ is constant on $P$, the exposed faces cancel; otherwise both have smaller dimension. Consequently,

$$D_u \mathcal{L}_k \subseteq \mathcal{L}_{k-1} \qquad (k \geq 1). \tag{46}$$

**The signed caterpillar sum.** We seek a combination that agrees with $h_\Delta$ up to a function in $\mathcal{L}_{N-2}$. Adding $x_{\pi(0)}$ makes the caterpillar function the following maximum

$$x_{\pi(0)} + U_{\pi,S}(x) = \max \left\{ x_{\pi(0)},\, x_{\pi(q+1)} + \sum_{\ell \in L_{\pi,S}} \sigma(\ell(x)) \right\}. \tag{47}$$

Averaging over permutations gives the required symmetry

$$F_N(x) := \frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,r\}} (-1)^{N+|S|} \big( x_{\pi(0)} + U_{\pi,S}(x) \big), \tag{48}$$

The signs cancel the extra exposed-face contributions modulo $\mathcal{L}_{N-3}$, as proved below.

**Theorem 14** (Dictionary sufficiency, `Lean_16`, Proof)**.** *For every $N \geq 3$, the dictionary of Definition 5 satisfies*

$$\max_N \in \operatorname{span} \mathcal{D}. \tag{49}$$

**Proof of Theorem 14.** Define the error by

$$f := F_N|_{H_0} - h_\Delta. \tag{50}$$

Step 14.1 reduces dictionary membership to $f \in \mathcal{L}_{N-2}$; Step 14.2 reduces this to $D_u f \in \mathcal{L}_{N-3}$ for all $u$. Step 14.3 proves these bounds by cancellation, using induction only when deleting a single exceptional coordinate.

**Step 14.1 (Reduce to a lower-dimensional error).** Assume temporarily that

$$f = F_N - h_\Delta \in \mathcal{L}_{N-2}. \tag{51}$$

Write $Q_{\pi,S} := Q_{h_{\pi,S}, L_{\pi,S}}$. Removing the linear apex terms from Equation (48) gives

$$\frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,N-2\}} (-1)^{N+|S|} h_{Q_{\pi,S}} - h_\Delta \in \mathcal{L}_{N-2}. \tag{52}$$

We correct the error by subtracting derivatives $\partial_u$ until it is linear. Faces of $Q_{\pi,S}$ already have support functions in the dictionary span: apex–base faces give Equation (14), base faces give linear functions plus hinges, and the apex alone gives zero. Coordinate features and Equation (8) represent the linear terms and either hinge orientation. Differentiating $h_\Delta$ also produces simplex faces; induction on their vertex count supplies these functions.

**Claim 14.1 [Simplex faces, `Lean_17`, `Lean_18`, Proof].** For every nonempty $T \subseteq \{0, e_1, \dots, e_{N-1}\}$, the support function of $\Delta_T := \operatorname{conv} T$ lies in the dictionary span.

**Start ($|T| = 1$).** The support function of a singleton is linear and lies in the dictionary span.

**Step ($|T| \geq 2$).** Fix $T$ and assume the claim for all smaller nonempty vertex sets. For any current vertex set $A$ with $|A| \geq 2$, remove $v \in A$ using a direction whose value at $v$ exceeds its common value at the other vertices. Writing $\Delta_A := \operatorname{conv} A$, this gives

$$\begin{aligned} u_v &:= \begin{cases} -\sum_{i=1}^{N-1} e_i & v = 0, \\ v & v \neq 0, \end{cases} \\ (\Delta_A)^{u_v} &= \{v\}, \qquad (\Delta_A)^{-u_v} = \Delta_{A\setminus\{v\}}, \\ D_{u_v} h_{\Delta_A}(x) &= \langle v, x\rangle - h_{\Delta_{A\setminus\{v\}}}(x). \end{aligned} \tag{53}$$

Order the vertices outside $T$ as $v_1, \dots, v_{N-|T|}$ and set

$$\varepsilon_T := D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} \left( \frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,N-2\}} (-1)^{N+|S|} h_{Q_{\pi,S}} - h_\Delta \right). \tag{54}$$

Each deletion changes the sign of the remaining simplex term, and subsequent $D_u$ annihilate the linear terms. Hence

$$D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} h_\Delta = (-1)^{N-|T|} h_{\Delta_T} + \ell_T, \tag{55}$$

where $\ell_T$ is linear. Thus the coefficient of $h_{\Delta_T}$ in $\varepsilon_T$ is $(-1)^{N-|T|+1}$; all other terms are linear or caterpillar-face support functions and lie in $\operatorname{span} \mathcal{D}$. By Equation (46), $\varepsilon_T \in \mathcal{L}_{|T|-2}$. For the full vertex set the empty composition is the identity.

Lower the residual dimension recursively by

$$\begin{aligned} \varepsilon_T^{(0)}(x) &:= \varepsilon_T(x), \\ \varepsilon_T^{(j+1)}(x) &:= \varepsilon_T^{(j)}(x) - \sum_{u \in E_{T,j}} \big(\partial_u \varepsilon_T^{(j)}\big)(x). \end{aligned} \tag{56}$$

Here $0 \leq j < |T| - 2$; the next claim chooses each finite $E_{T,j} \subseteq H_0$.

**Claim 14.2 [Residual reduction, `Lean_19`, Proof].** The sets $E_{T,j}$ can be chosen successively so that, for $0 \leq j \leq |T| - 2$,

$$\begin{aligned} \varepsilon_T^{(j)} &\in \mathcal{L}_{|T|-2-j}, \\ \varepsilon_T - \varepsilon_T^{(j)} &\in \operatorname{span} \mathcal{D}. \end{aligned} \tag{57}$$

Each residual remains a linear combination of $h_{\Delta_T}$, support functions of its proper faces and of caterpillar faces, and linear functions.
The proof in Appendix A.1 chooses a direction orthogonal to each maximal-dimensional direction space, avoiding orthogonality to the simplex and other spaces. The derivative sum preserves each maximal term exactly once; subtracting it lowers the dimension.

At stage $j = |T| - 2$, the error is linear, so Equation (57) gives $\varepsilon_T \in \operatorname{span} \mathcal{D}$. Its simplex coefficient is $\pm 1$, and its other terms already lie in the span; hence $h_{\Delta_T} \in \operatorname{span} \mathcal{D}$. Taking the full vertex set proves $h_\Delta \in \operatorname{span} \mathcal{D}$ under assumption Equation (51).

**Step 14.2 (Why the facet bounds imply the class identity).** For $N \geq 4$ and $f = F_N - h_\Delta$, we show that $f \in \mathcal{L}_{N-2}$ follows from

$$D_u f \in \mathcal{L}_{N-3} \qquad \text{for every } u \in H_0. \tag{58}$$

The base zonotope admits the Freudenthal–Kuhn triangulation [Fre42, Kuh60]; coning it to the apex expresses each caterpillar support function as a sum of simplex support functions modulo $\mathcal{L}_{N-2}$. For a simplex, $D_u$ isolates a facet up to the linear support function of its opposite vertex. Thus Equation (58) controls the weighted sums of parallel facets. A common linear reconstruction operator turns these facet cancellations back into a cancellation of the original simplex sum. Define

$$(\mathcal{T} g)(x) := \begin{cases} g(x) & x_1 \geq 0, \\ g(x - x_1 e_1) & x_1 < 0. \end{cases} \tag{59}$$

**Claim 14.3 [Simplex reconstruction, `Lean_20`, Proof].** Every $(N-1)$-dimensional simplex $P \subseteq H_0$ satisfies

$$h_P - \mathcal{T}\left( \sum_{F \text{ facet of } P} \operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F \right) \in \mathcal{L}_{N-2}. \tag{60}$$

Here $\nu_F$ is an outward normal of $F$. Its sign against $e_1$ records whether the facet bounds lines parallel to $e_1$ from above or below; parallel facets contribute zero. The operator $\mathcal{T}$ fills the segments between a polytope and its projection onto $p_1 = 0$, up to a segment correction for translation; hence $\mathcal{T}\mathcal{L}_k \subseteq \mathcal{L}_{k+1}$. Applying the claim with the same operator to every simplex preserves the cancellations between parallel facets. Their sums lie in $\mathcal{L}_{N-3}$, so their reconstructions lie in $\mathcal{L}_{N-2}$, as required. Appendix A.2 proves the claim and the subdivision and grouping steps.

**Claim 14.4 [Class identity, `Lean_21`, Proof].** The class identity Equation (51) holds for every $N \geq 3$.

**Step 14.3 (Prove the class identity by induction).**

**Start ($N = 3$).** Here $S = \{1\}$, $q = 1$, and $p(1) = 0$, so Equation (47) gives

$$\begin{aligned} x_{\pi(0)} + U_{\pi,\{1\}}(x) &= \max\bigl\{ x_{\pi(0)}, x_{\pi(2)} + \sigma(x_{\pi(1)} - x_{\pi(2)}) \bigr\} \\ &= \max\{ x_{\pi(0)}, x_{\pi(1)}, x_{\pi(2)} \} = \max_3(x). \end{aligned} \tag{61}$$

All six terms have coefficient $1/6$, so

$$F_3 = \max_3 . \tag{62}$$

On $H_0$, this gives $f = 0 \in \mathcal{L}_1$.

**Step ($N \geq 4$).** Assume the class identity for $N-1$. By Step 14.2, it suffices to prove $D_u f \in \mathcal{L}_{N-3}$ for every $u \in H_0$. The linear apex terms disappear. The exposed simplex faces have the largest and smallest coordinate values $u_i$, including $u_0 = 0$. In a caterpillar base, $[0, \ell_{ij}]$ remains free when $u_i = u_j$; its maximizing endpoint is $\ell_{ij}$ when $u_i > u_j$ and 0 when $u_i < u_j$. Minimization reverses these choices.

**Which faces can survive?** Fix $u \in H_0 \setminus \{0\}$; the direction 0 gives $D_0 f = 0$. A base face is a zonotope, whose support function is linear plus a sum of segment support functions, hence belongs to $\mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. The apex alone contributes zero. If $j$ tree edges have unequal endpoint values under $u$, an apex–base face has at most $N-2-j$ free segments and therefore dimension at most $N-1-j$. Thus only $j \leq 1$ needs attention.

Write $a = u_{\pi(0)}$. An apex–base face requires the maximum over the base to be zero. For $j = 0$, all tree values equal $c$, so this condition is $c - a = 0$, forcing $u = 0$. For $j = 1$, deleting the unequal edge leaves constant values $c \neq d$ on its two components. The zero-maximum condition is

$$a = \max\{c, d\} \qquad \text{for a spine edge,} \tag{63}$$

$$a = c + \sigma(c - d) \qquad \text{for a leaf edge, with leaf value } d. \tag{64}$$

The spine case gives two coordinate values. The leaf case does too when $d > c$; when $d < c$, it gives the values $c + t, c - t, c, \ldots, c$, where $t = c - d > 0$. Replacing $u$ by $-u$ treats the opposite face. Consequently, only two-value directions and these symmetric three-value directions can contribute modulo $\mathcal{L}_{N-3}$.

**Three values: reflection symmetry.** Suppose $u_\alpha = c + t$, $u_\beta = c - t$, and all other coordinates equal $c$. Let $\tau$ exchange $\alpha$ and $\beta$. On $H_0$, its coordinate action is

$$(Rx)_i = x_{\tau(i)} - x_{\tau(0)}, \qquad Rx = x - \lambda(x)u, \qquad \lambda(x) := \frac{x_\alpha - x_\beta}{t}. \tag{65}$$

In particular $Ru = -u$. For $G = F_N|_{H_0}$, permutation symmetry and a common coordinate shift give $G \circ R = G + \ell$ with $\ell$ linear. Every exposed face in direction $v$ lies in the hyperplane $\langle v, p \rangle = h_P(v)$; hence $\partial_v G(x + sv) = \partial_v G(x) + sG(v)$. Applying this to the derivative of $G \circ R = G + \ell$ at $u$ gives

$$\partial_u G(x) + \ell(x) = \partial_{-u} G(Rx) = \partial_{-u} G(x) + \lambda(x) G(-u). \tag{66}$$

Therefore

$$D_u G = \lambda\, G(-u) - \ell \in \mathcal{L}_0. \tag{67}$$

The two exposed simplex faces are vertices, so $D_u f \in \mathcal{L}_0$ as well.

**Handling two distinct values.** First suppose that each value occurs at least twice. After the preceding dimension bound, every term still requiring cancellation has a unique split $u_{\pi(1)} = \cdots = u_{\pi(k)} \neq u_{\pi(k+1)} = \cdots = u_{\pi(r+1)}$ with $k \in S$. If the apex label has the latter value, toggle membership of $k-1$ in $S$. Otherwise, exchange the equal-valued labels $\pi(k+1), \pi(k+2)$ and toggle membership of $k+1$. Repetition of both values gives $k \geq 2$ in the first case and $k + 2 \leq r + 1$ in the second, so the operations are defined; the unchanged cut at $k$ keeps the cut set nonempty. The transformed term remains in the same case, and repeating the operation returns the original term. If $\Phi$ denotes this involution on the active terms and $w_{\pi,S} := (-1)^{N+|S|}/N!$, then

$$w_{\Phi(\pi,S)} = -w_{\pi,S}, \qquad D_u h_{Q_{\pi,S}} - D_u h_{Q_{\Phi(\pi,S)}} \in \mathcal{L}_1. \tag{68}$$

Inactive fixed points of the global transformation are already in $\mathcal{L}_{N-3}$ by the dimension bound. Summing the active pairs therefore puts $D_u F_N$ in $\mathcal{L}_{N-3}$. Each exposed simplex face has at most $N-2$ vertices and hence dimension at most $N-3$, so $D_u f \in \mathcal{L}_{N-3}$ without using induction.

**One exceptional coordinate: the induction hypothesis.** Suppose one of the two values occurs only once. Choose $\tau$ with $\tau(0)$ the exceptional label. To stay in $H_0$, relabel and subtract the new zeroth coordinate:

$$B_\tau : H_0 \longrightarrow H_0, \qquad (B_\tau x)_i := x_{\tau(i)} - x_{\tau(0)} \qquad (0 \le i < N). \tag{69}$$

Its inverse is $B_{\tau^{-1}}$, and it sends $u$ to $(0, t, \dots, t)$, $t \neq 0$. Transforming both input and direction preserves every $\mathcal{L}_k$ and changes $f$ only by a linear term, annihilated by $D$. Rescaling, and reversing the sign of $D_u$ if needed, reduces to $u = (0, 1, \dots, 1)$.

This direction exposes the $(N-2)$-dimensional facet $\Delta' := \mathrm{conv}\{e_1, \dots, e_{N-1}\}$ and the opposite vertex $0$. Hence

$$D_u h_\Delta(x) = h_{\Delta'}(x) = \max\{x_1, \dots, x_{N-1}\} \qquad (x \in H_0). \tag{70}$$

The induction hypothesis also applies to unrestricted lower-arity inputs modulo linear functions. Indeed, put $z = (x_1, \dots, x_{N-1})$ and $y = z - x_1\mathbf{1}$. Each summand of $F_{N-1}$ increases by the common shift, so

$$F_{N-1}(z) - \max_{N-1}(z) \equiv F_{N-1}(y) - \max_{N-1}(y) \pmod{\mathcal{L}_0}. \tag{71}$$

The first coordinate of $y$ is zero. Induction and linear precomposition therefore give

$$\big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big) \in \mathcal{L}_{N-3}. \tag{72}$$

**Claim 14.5 [Coordinate deletion, `Lean_22`, Proof].** For $N \ge 4$,

$$D_{(0,1,\dots,1)}(F_N|_{H_0}) - \big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1})\big) \in \mathcal{L}_{N-3}. \tag{73}$$

The proof is in Appendix A.3. Its mechanism is that only degree-one tree positions of label $0$ survive. Each nonempty lower cut set has $N$ insertion preimages, each contributing $1/N$ of its lower coefficient. The extra empty-cut star sum lies in $\mathcal{L}_{N-3}$ by Boolean cancellation.

**Completion of the induction.** Combining Equation (70), Equation (72), and Claim 14.5 gives

$$\begin{aligned} D_u f(x) = {} & \big[D_u(F_N|_{H_0})(x) - F_{N-1}(x_1, \dots, x_{N-1})\big] \\ & + \big[F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big]. \end{aligned} \tag{74}$$

Both brackets lie in $\mathcal{L}_{N-3}$, the first by Claim 14.5 and the second by induction. Step 14.2 now gives $f \in \mathcal{L}_{N-2}$, and Step 14.1 gives the dictionary representation. □

# 6 Network size

The widths in Theorem 1 are worst-case upper bounds: we count all possible descriptions after collecting linear terms into the head and orienting the remaining hinges consistently. Discarding zero aggregate coefficients can give a smaller network; zero padding realizes the stated widths.

The normalized description Equation (21) of a caterpillar function depends only on its undirected spine, apex, and leaf attachments. For $m$ spine vertices these choices give the summand of $C_N$ in Equation (6). Each of the $N-2$ generators has three face states, giving at most $3^{N-2}C_N$ descriptions. The $N(N-1)$ descriptions with no retained hinge and head $x_i - x_j$, $i \neq j$, can be combined with the auxiliary functions. By Theorem 6, the second width is the number of retained descriptions plus two; Appendix B gives the complete count.

**Formalization (`Lean_23`).** In this working draft, the theorem and equation links point to the local Lean sources. Appendix E prints the network definitions and the main theorem, and describes the verification certificate.

# 7 Compression

Theorem 6 reduces the second width to the number of retained units in an expansion. This section bounds that number through the dimension of the vocabulary. Let $V_N := \operatorname{span} \mathcal{D}$, and let $A_N \subseteq V_N$ be the span of the auxiliary functions Equation (18). Put

$$q_N := \dim\big(V_N/A_N\big) = \dim V_N - \dim A_N. \tag{75}$$

**Theorem 15** (Quotient compression, `Lean_24`, Proof)**.** *For every $N \geq 3$ there are a set $K \subseteq \mathcal{K}_0$ with $|K| \leq q_N$, rational coefficients $d_k$, and $R \in A_N$ with rational auxiliary coefficients such that $\max_N = \sum_{k\in K} d_k U_k + R$. Hence $\max_N$ is computed by a network with rational weights, zero biases, and widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = q_N + 2. \tag{76}$$

**Proof of Theorem 15.** Choose a basis of $A_N$ among the auxiliary functions and extend it by face functions to a basis of $V_N$. The extension adds exactly $q_N$ face functions. For each of them select one normalized description, rather than splitting its coefficient across all descriptions of the same function; this gives $K$ with $|K| \leq q_N$. Let $S = \{g_1, \dots, g_m\}$ be the resulting basis of $V_N$. By Theorem 14, $\max_N \in V_N$, so $\max_N = \sum_i c_i g_i$ with unique real coordinates $c_i$.

The coefficients can be made rational without changing the support. Choose a $\mathbb{Q}$-linear map $P : \mathbb{R} \to \mathbb{Q}$ fixing $\mathbb{Q}$ by extending $\{1\}$ to a $\mathbb{Q}$-basis of $\mathbb{R}$. Every $g_i$ and $\max_N$ is continuous and rational-valued on $\mathbb{Q}^N$. For $y \in \mathbb{Q}^N$, applying $P$ to the expansion gives

$$\max_N(y) = P\left(\sum_i c_i g_i(y)\right) = \sum_i P(c_i) g_i(y). \tag{77}$$

Continuity extends the equality to all of $\mathbb{R}^N$, with rational coefficients $P(c_i)$. Theorem 6 compiles the expansion; its weights are rational because the new coefficients are rational. □

Every face function is the unit of a normalized description outside $\mathcal{P}$ or an auxiliary combination (Appendix B), so $q_N + 2$ never exceeds the second width of Theorem 1. This dimension controls compression by a basis of the whole vocabulary. An individual expansion of $\max_N$ may use fewer functions.

# 8 CPWL functions

By the Wang–Sun theorem, every CPWL function on $\mathbb{R}^d$ is a signed sum of maxima of $d+1$ affine functions. These affine functions can be taken from any finite family among which the function selects pointwise, without rescaling them.

**Theorem** (Koutschan–Moser–Ponomarchuk–Schicho, [KMPS23, Theorem 2.4], `Lean_25`, `Lean_26`)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous, and let $g_1, \dots, g_n : \mathbb{R}^d \to \mathbb{R}$ be affine with $f(x) \in \{g_1(x), \dots, g_n(x)\}$ for every $x$. Then there are $T \in \mathbb{N}$, signs $\varepsilon_t \in \{-1, 1\}$, and maps $\alpha_t : \{0, \dots, d\} \to \{1, \dots, n\}$ with*

$$f(x) = \sum_{t=1}^{T} \varepsilon_t \max\nolimits_{d+1}\big(g_{\alpha_t(0)}(x), \dots, g_{\alpha_t(d)}(x)\big) \qquad \textit{for every } x \in \mathbb{R}^d. \tag{78}$$

The affine functions of every polyhedral description of a CPWL function satisfy the hypothesis. The Lean proof follows Ovchinnikov's max–min form $f = \max_S \min_{j \in S} g_j$, expands it by inclusion–exclusion into an integer combination of maxima over subsets of $\{g_1, \ldots, g_n\}$, and reduces each maximum to subsets of at most $d+1$ functions by an affine dependence among $d+2$ gradients.

**Reusing the first hidden layer (`Lean_27`, `Lean_28`).** Let $d \geq 2$, and let $(h_1, h_2)$ be the widths of Theorem 1 for $N = d+1$. Substituting the arguments of each term of Equation (78) into the Caterpillar Net computes $f$. All terms use only the features $\sigma(g_a - g_b)$ for $a < b$, $\sigma(\pm g_1)$, and $\sigma(g_a - g_1)$, so one first hidden layer of width $\binom{n}{2} + n + 1$ serves every term. Terms with the same argument set $\alpha_t(\{0, \ldots, d\})$ compute the same maximum and are merged, and the auxiliary remainders of all terms form one pair $\sigma(\pm R)$. The second hidden layer has width $s(h_2 - 2) + 2$, where $s$ is the number of distinct argument sets and $s \leq \min\bigl(T, \sum_{m \leq d+1} \binom{n}{m}\bigr)$. Thus the second hidden layer holds one copy of the Caterpillar Net units for every distinct maximum; only the first hidden layer and the remainder pair are genuinely shared. The second hidden layer and output layer have rational weights and zero biases, and every first hidden layer neuron applies $\sigma$ to $\pm(g_a - g_b)$ or $\pm g_a$. Hence rational $g_i$ give rational weights and biases, and linear $g_i$ give zero biases.

# 9 Conclusion

The second width of the Caterpillar Net is the number of caterpillar functions in an expansion of $\max_N$ plus two. For general $N$, Theorem 15 retains at most $q_N$ of them. Conversely, every such expansion needs at least $(7/6)^N/(N+1)$ caterpillar functions (Theorem 8), so caterpillar building blocks cannot give subexponential second widths. Reducing our upper bound $2^{O(N \log N)}$ to $2^{O(N)}$ remains open. The factorial dimension bound in Theorem 13 limits compression by a basis of the entire dictionary; it does not give a factorial lower bound for representing the single function $\max_N$. Whether the maximum has two-hidden-layer representations of subexponential width with other second-layer functions remains open.

# A Supporting proofs for the dictionary theorem

## A.1 Residual reduction

We prove the inner induction used in the simplex-face argument of Section 5. Fix a nonempty vertex set $T$ with $|T| \geq 2$, and assume that every proper simplex face on a subset of $T$ is already in the dictionary span, as in the outer induction there.

**Proof of Claim 14.2.** We argue by induction on $j$.

**Start ($j = 0$).** By definition, $\varepsilon_T^{(0)} = \varepsilon_T$. Its dimension bound and support-function expression were established above, and the removed part is zero.

**Step ($j < |T| - 2$).** Assume the directions for the first $j$ stages have been chosen and the claim holds at stage $j$. Put $k := |T| - 2 - j \geq 1$. We require each new direction $u$ to be nonorthogonal to the simplex direction space $\operatorname{dir} \Delta_T := \operatorname{span}\{v - w : v, w \in T\}$. Its values on $T$ are then not all equal, so the maximizing vertices form a nonempty proper subset:

$$\begin{aligned} A_u &:= \{v \in T : \langle u, v\rangle = h_{\Delta_T}(u)\}, \qquad \varnothing \neq A_u \subsetneq T, \\ \partial_u h_{\Delta_T} &= h_{\Delta_{A_u}} \in \operatorname{span} \mathcal{D}. \end{aligned} \tag{79}$$

The last membership uses the outer induction hypothesis, since $|A_u| < |T|$. An orthogonal direction would leave $h_{\Delta_T}$ unchanged and would not allow this use of the hypothesis.

Choose an auxiliary expression for $\varepsilon_T^{(j)}$ by support functions of polytopes of dimension at most $k$, and collect their distinct $k$-dimensional direction spaces. For each such space $V$, choose $u_V \in V^\perp$ not orthogonal to $\operatorname{dir} \Delta_T$ or to any other collected space $V'$. This is possible because $\dim \Delta_T = |T| - 1 > k$: each of $\operatorname{dir} \Delta_T$ and $V'$ contains a vector outside $V$, whose nonzero projection onto $V^\perp$ is not orthogonal to that vector. Adding sufficiently small multiples of these projections avoids each orthogonality condition in turn while preserving those already avoided. Choose $E_{T,j} := \{u_V : V \text{ is a collected direction space}\}$, with $E_{T,j} = \varnothing$ if there are none. In the sum of derivatives over $E_{T,j}$, each term coming from a $k$-dimensional polytope is preserved exactly once. Every other derivative of that term is the support function of a proper face, while terms of lower dimension remain in $\mathcal{L}_{k-1}$. Subtracting this derivative sum therefore cancels all $k$-dimensional terms. By the recursive definition Equation (56), the resulting error satisfies $\varepsilon_T^{(j+1)} \in \mathcal{L}_{k-1} = \mathcal{L}_{|T|-3-j}$.

Now compute these same derivatives using the support-function expression in the claim. The derivative of $h_{\Delta_T}$ lies in the dictionary span by Equation (79). Differentiating the support function of an already proper simplex face still gives a support function of a proper face of $\Delta_T$, so the same induction hypothesis applies. Differentiating the support function of a face of $Q_{\pi,S}$ gives a support function of a face of the same polytope, whose support functions lie in the dictionary span, and linear functions remain linear. Thus the derivative sum lies in the dictionary span. By Equation (56), $\varepsilon_T - \varepsilon_T^{(j+1)}$ is the sum of the previously removed part and this derivative sum, so it also lies in the dictionary span. Subtracting the derivative sum preserves the stated types of support-function terms. This completes the proof step.

□

## A.2 Reconstruction from opposite exposed faces

We justify Step 14.2: for $N \geq 4$, the bounds Equation (58) imply $f \in \mathcal{L}_{N-2}$. We first subdivide the caterpillar polytopes, then prove Claim 14.3, and finally apply it to the resulting weighted simplex sum.

The base of each $Q_{\pi,S}$ is an affine cube. Order its edge vectors by a bijection $\omega : \{1, \dots, N-2\} \to L_{\pi,S}$, add them one at a time, and adjoin the apex:

$$K_{\pi,S,\omega} := \operatorname{conv}\left(\{0\} \cup \left\{h_{\pi,S} + \sum_{a=1}^{k} \omega(a) : 0 \le k \le N-2\right\}\right). \tag{80}$$

These $(N-2)!$ simplices are obtained by coning the *Freudenthal–Kuhn triangulation* of the base to the apex [Fre42, Kuh60]: in cube coordinates, each base simplex occupies one ordering region $1 \ge t_{\omega(1)} \ge \cdots \ge t_{\omega(N-2)} \ge 0$. For a hyperplane cut $P = P_+ \cup P_-$, the support-function identity

$$h_P + h_{P_+ \cap P_-} = h_{P_+} + h_{P_-} \tag{81}$$

shows that subdivision changes the sum of support functions only by support functions of the cutting sections. Hence

$$h_{Q_{\pi,S}} - \sum_{\omega} h_{K_{\pi,S,\omega}} \in \mathcal{L}_{N-2}. \tag{82}$$

Substituting Equation (82) into Equation (48) expresses $f$ as a finite linear combination of support functions of $(N-1)$-dimensional simplices in $H_0$, plus a remainder in $\mathcal{L}_{N-2}$. The simplex $\Delta$ contributes its support function with coefficient $-1$, and the linear apex terms are included in the remainder.

For a simplex $P$, an outward facet normal $\nu_F$ exposes the facet $F$ on one side and its opposite vertex $v_F$ on the other. Thus

$$D_{\nu_F} h_P(x) = h_F(x) - \langle v_F, x \rangle. \tag{83}$$

Directions exposing no facet on either side give support functions of faces of dimension at most $N-3$. Consequently, Equation (58) says that each weighted, signed sum of parallel-facet support functions belongs to $\mathcal{L}_{N-3}$.

Our assumption controls the weighted facet contributions, whereas our conclusion concerns $f$. To recover the original sum modulo $\mathcal{L}_{N-2}$, we use the same linear operator for every simplex, so cancellations between facet contributions survive reconstruction. Use the operator $\mathcal{T}$ defined in Equation (59). The sign in this definition concerns the input $x$. To interpret it geometrically, first suppose that every point $p \in P$ satisfies $p_1 \ge 0$, and write $\operatorname{pr}(p) := p - p_1 e_1$ for projection onto $p_1 = 0$. Since $\langle p, x \rangle - \langle \operatorname{pr}(p), x \rangle = p_1 x_1$, each point dominates its projection when $x_1 \ge 0$, and its projection dominates when $x_1 < 0$. Consequently,

$$\mathcal{T} h_P(x) = \max\{h_P(x), h_{\operatorname{pr} P}(x)\} = h_{\operatorname{conv}(P \cup \operatorname{pr} P)}(x). \tag{84}$$

This convex hull fills the segments with endpoints $p$ and its projection, for $p \in P$, so its dimension is at most $\dim P + 1$. For an arbitrary $P$, choose $c \ge 0$ such that $P + ce_1$ lies in $p_1 \ge 0$. Translation adds $cx_1$ to the support function, and $\mathcal{T}(x_1) = \sigma(x_1)$, giving

$$\mathcal{T} h_P(x) = h_{\operatorname{conv}((P+ce_1) \cup \operatorname{pr}(P+ce_1))}(x) - c\sigma(x_1). \tag{85}$$

The correction $\sigma(x_1)$ is the support function of the segment $[0, e_1]$. Thus linearity gives

$$\mathcal{T} \mathcal{L}_k \subseteq \mathcal{L}_{k+1}. \tag{86}$$

**Proof of Claim 14.3.** Let $k = |\{F \text{ facet of } P : \langle \nu_F, e_1 \rangle \ne 0\}|$ count facets not parallel to $e_1$. Boundedness ensures both nonzero signs occur, so $k \ge 2$.

**Start ($k = 2$).** There is one upper facet $F^+$ and one lower facet $F^-$. Every line parallel to $e_1$ that meets $P$ has its two intersection endpoints on these facets. Thus $h_P(x) = h_{F^+}(x)$ when $x_1 \geq 0$, and $h_P(x) = h_{F^-}(x)$ when $x_1 < 0$. When $x_1 \geq 0$, $\mathcal{T}$ evaluates at $x$, so subtracting $\mathcal{T}(h_{F^+} - h_{F^-})$ from $h_P$ leaves $h_{F^-}$. When $x_1 < 0$, $\mathcal{T}$ evaluates both facet support functions at $x - x_1 e_1$, where they agree because the facets have the same projection. Their difference is therefore zero, again leaving $h_{F^-}$. The reconstruction error is therefore $h_{F^-} \in \mathcal{L}_{N-2}$, proving the claim for $k = 2$.

**Step ($k > 2$).** Assume the claim for $k - 1$, and let $P$ have $k$ facets not parallel to $e_1$. Two of them have the same nonzero sign of $\langle \nu_F, e_1 \rangle$. Their intersection is an $(N-3)$-dimensional face $C$. Let $W := \operatorname{span}\{p - q : p, q \in C\}$, and project orthogonally within $H_0$ onto $W^\perp$. This is a two-dimensional plane, since $\dim H_0 = N - 1$ and $\dim W = N - 3$. All points of $C$ have the same projection $c$, because their differences lie in $W$. There are just two vertices of $P$ outside $C$, so the projected simplex is a triangle with vertices $c$ and the projections of those two vertices. The chosen facets project to its two sides meeting at $c$.

The chosen facet normals are orthogonal to $W$, so their signs against $e_1$ equal their signs against its projection. This projected direction is nonzero, since the chosen facets are not parallel to $e_1$. If both sides are upper, moving from $c$ opposite to this direction enters the triangle; if both are lower, moving in this direction enters it. Continuing meets the interior of the opposite edge. The inverse image of the line through $c$ in this direction is a hyperplane in $H_0$ containing $C$ and parallel to $e_1$. It cuts the edge whose endpoints are the two vertices outside $C$ at an interior point and divides $P$ into two $(N-1)$-dimensional simplices. Each keeps one of the two chosen facets and replaces the other by the new facet parallel to $e_1$. Its other facets retain their normal directions, so each new simplex has exactly $k - 1$ facets not parallel to $e_1$.

Write $E(Q)$ for the reconstruction error on the left of Equation (60). Let $P_+$ and $P_-$ be the two new simplices, let $R := P_+ \cap P_-$ be their common cutting facet, and write $F_1, F_2$ for the chosen facets and $R_F := R \cap F$ for every other facet of $P$. With $s_F := \operatorname{sgn}\langle \nu_F, e_1 \rangle$, applying Equation (81) to $P$ and to every split facet gives the exact residual identity

$$E(P) = E(P_+) + E(P_-) - h_R + \mathcal{T}\left( \sum_{\substack{F \text{ facet of } P \\ F \notin \{F_1, F_2\}}} s_F h_{R_F} \right). \tag{87}$$

Indeed, the inherited facets retain their signs, while the two copies of $R$ are parallel to $e_1$ and therefore have sign zero. By the induction hypothesis, $E(P_+), E(P_-) \in \mathcal{L}_{N-2}$; moreover, $\dim R = N - 2$ and $\dim R_F \leq N - 3$. Equation Equation (86) therefore puts every term on the right of Equation (87) in $\mathcal{L}_{N-2}$. This proves the claim for $k$ and completes the induction.

□

**Conclusion for $f$.** Apply Equation (60) to every simplex support-function term in the expansion of $f$, keeping its original coefficient. A facet $F$ parallel to $e_1$ has $\langle \nu_F, e_1 \rangle = 0$, so its term in the reconstruction formula is zero and can be omitted. For each remaining occurrence of a facet, choose $\nu_F$ outward from its simplex and put

$$u_F := \frac{\nu_F}{\langle \nu_F, e_1 \rangle}, \qquad \langle u_F, e_1 \rangle = 1. \tag{88}$$

Thus the coordinate with index 1 is normalized to 1; the coordinate with index 0 remains zero because $u_F \in H_0$. Oppositely oriented parallel normals give the same $u_F$. Only the normal is rescaled: the support function and its coefficient are unchanged.

Let $\mathcal{U}$ be the finite set of vectors $u_F$ obtained this way. For each $u \in \mathcal{U}$, define $G_u$ by adding all facet support-function terms with $u_F = u$, retaining their signs and weights. An occurrence of $F$ as a

facet of $K_{\pi,S,\omega}$ contributes $(-1)^{N+|S|}\operatorname{sgn}\langle \nu_F, e_1\rangle\, h_F/N!$, while an occurrence as a facet of $\Delta$ contributes $-\operatorname{sgn}\langle \nu_F, e_1\rangle\, h_F$. These are precisely the terms with the same normalized normal, including all repetitions.

More explicitly, define $\rho$ to be the sum of the linear apex terms and the weighted subdivision errors in Equation (82). Then

$$f = \frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}\sum_{\omega} h_{K_{\pi,S,\omega}} - h_\Delta + \rho, \qquad \rho\in\mathcal{L}_{N-2}. \tag{89}$$

For a simplex $P$, collect its facets with normalized normal $u$ as

$$B_u(P) := \sum_{\substack{F \text{ facet of } P\\ \langle \nu_F, e_1\rangle\neq 0,\ u_F = u}} \operatorname{sgn}\langle \nu_F, e_1\rangle h_F. \tag{90}$$

An empty sum is zero. If $\langle \nu_F, e_1\rangle > 0$, the direction $u_F$ exposes $F$; if it is negative, $-u_F$ exposes $F$. The opposite face is a single vertex, so Equation (83) gives the signed facet term up to a linear function. If neither $u$ nor $-u$ exposes a facet of $P$, both exposed faces have dimension at most $N-3$. Thus $D_u h_P - B_u(P)\in\mathcal{L}_{N-3}$. Applying this to the full weighted expansion, with the same permutations and nonempty cut sets as in Equation (48), gives

$$\begin{aligned} D_u f - G_u = &\sum_{\pi,S\neq\varnothing}\frac{(-1)^{N+|S|}}{N!}\sum_{\omega}\big(D_u h_{K_{\pi,S,\omega}} - B_u(K_{\pi,S,\omega})\big)\\ &- (D_u h_\Delta - B_u(\Delta)) + D_u\rho \in \mathcal{L}_{N-3} \qquad (u\in\mathcal{U}). \end{aligned} \tag{91}$$

Together with Equation (58), relation Equation (91) puts every $G_u$ in $\mathcal{L}_{N-3}$. By linearity, the sum of the reconstructed facet terms is $\mathcal{T}(\sum_{u\in\mathcal{U}} G_u)$, which lies in $\mathcal{L}_{N-2}$ by Equation (86). Its difference from $f$ consists of the reconstruction and subdivision remainders, also in $\mathcal{L}_{N-2}$. Hence $f\in\mathcal{L}_{N-2}$. This proves the implication used in Step 14.2.

### A.3 Coordinate deletion

**Proof of Claim 14.5.** We now determine which terms of the signed sum remain for $u = (0,1,\dots,1)$. Only label $0$ has a different coordinate value from the others. Consequently, a tree-edge generator is nonorthogonal to $u$ exactly when one of its endpoint labels is $0$.

If label $0$ is the apex label, all tree-edge generators are orthogonal to $u$. The maximizing face is the whole base $B_{\pi,S}$ and the minimizing face is $\{0\}$, so $D_u h_{Q_{\pi,S}} = h_{B_{\pi,S}} \in \mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. If label $0$ is a tree vertex of degree at least two, at least two segment generators are nonorthogonal to $u$. The preceding dimension bound then gives $D_u h_{Q_{\pi,S}}\in\mathcal{L}_{N-3}$ as well. We can therefore discard both kinds of term: it suffices to retain descriptions in which label $0$ is a tree vertex of degree one.

To express this condition in the permutation notation, let $p := \pi^{-1}(0)$ be the index of label $0$ in the full order $(\pi(0),\dots,\pi(N-1))$. When $p\geq 1$, its tree position is $p-1$. A degree-one vertex is either an endpoint of the spine with no attached leaves or a leaf outside the spine. The retained descriptions are thus exactly

$$\begin{aligned} &p = 1, \quad 1\in S && \text{(root endpoint)},\\ &2\leq p\leq N-1, \quad p-1\notin S && \text{(leaf outside the spine)},\\ &p = N-1, \quad N-2\in S && \text{(last spine endpoint)}. \end{aligned} \tag{92}$$

Indeed, the root has only one child precisely when $1\in S$; a nonroot spine vertex has a parent and also a child unless it occupies the final tree position. Deleting label $0$ from the full order now gives an order $\bar\pi$ mapping positions $\{0,\dots,N-2\}$ bijectively to labels $\{1,\dots,N-1\}$.

For $0 \le i \le N-3$, let $\iota_i$ insert a gap at the prospective pendant position $i+1$. Given a lower cut set $T \subseteq \{1,\dots,N-3\}$, the three active insertion rules are

$$\begin{aligned} \iota_i(t) &:= \begin{cases} t, & t < i+1, \\ t+1, & t \ge i+1, \end{cases} \\ \text{first end:} \quad & (p,S) = (1, \{1\} \cup \{t+1 : t \in T\}), \\ \text{pendant } i\text{:} \quad & (p,S) = (i+2, \{\iota_i(t) : t \in T\}) \qquad (0 \le i \le N-3), \\ \text{last end:} \quad & (p,S) = (N-1, T \cup \{N-2\}). \end{aligned} \tag{93}$$

In every row the upper permutation is uniquely obtained by inserting label $0$ at position $p$ in $\bar\pi$. Conversely, with $t$ restricted to $\{1,\dots,N-3\}$, the deletion rules are $T = \{t : t+1 \in S\}$ for the first end, $T = \{t : \iota_i(t) \in S\}$ for pendant $i$, and $T = S \setminus \{N-2\}$ for the last end. These operations are inverse. The three cases in Equation (92) correspond respectively to the first-end row, the pendant row with $i = p-2$, and the last-end row. Thus a nonempty $T$ has $N-2$ pendant and two endpoint preimages, whereas $T = \varnothing$ has only the two endpoints because upper cut sets are nonempty.

Extend the lower caterpillar notation to $T = \varnothing$ by setting its last spine position and every parent equal to $0$, so its head begins at the root $\bar\pi(1)$. For every lower label order $\bar\pi$ put

$$\begin{aligned} C_{\bar\pi,T}(x) &:= x_{\bar\pi(0)} + U_{\bar\pi,T}(x), \\ q_T &:= \max(\{0\} \cup T), \qquad p_T(j) := \max(\{0\} \cup \{t \in T : t < j\}). \end{aligned} \tag{94}$$

For each upper preimage $(\pi, S)$ of $(\bar\pi, T)$, the insertion identities and the literal factorial coefficients are

$$\begin{aligned} \text{pendant:} \quad & \frac{(-1)^{N+|S|}}{N!} = -\frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \quad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) + C_{\bar\pi,T} \in \mathcal{L}_1, \\ \text{endpoint:} \quad & \frac{(-1)^{N+|S|}}{N!} = \frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \qquad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) - C_{\bar\pi,T} \in \mathcal{L}_1. \end{aligned} \tag{95}$$

Thus every one of the $N-2+2 = N$ preimages contributes $1/N$ of the lower summand modulo $\mathcal{L}_{N-3}$, and together they recover its coefficient in $F_{N-1}$.

The extra terms are the two endpoint fibres whose deletion leaves $T = \varnothing$, hence a rooted star. By Equation (95), each contributes $1/N$ of the weighted lower summand modulo $\mathcal{L}_1$. For a fixed lower order $\bar\pi$, the nonroot positions are $J = \{0, 2, \dots, N-2\}$, including the original apex position $0$. For $a \in J$, let $\bar\pi^{(a)}$ exchange positions $0$ and $a$, keeping the root at position $1$; the exchange at $a = 0$ is the identity. Put

$$\begin{aligned} q := |J| = N-2, \qquad & d_a := x_{\bar\pi(1)} - x_{\bar\pi(a)} \quad (a \in J), \\ T_A := \sigma\left(\sum_{a\in A} d_a + \sum_{a \in J\setminus A} \sigma(d_a)\right) \quad & (A \subseteq J), \\ C_{\bar\pi^{(a)},\varnothing} = x_{\bar\pi(a)} + T_{\{a\}} \quad & (a \in J). \end{aligned} \tag{96}$$

The Boolean alternating identity gives

$$\sum_{A\subseteq J} (-1)^{|A|} T_A = 0 \quad \Longrightarrow \quad \sum_{a\in J} T_{\{a\}} = T_\varnothing + \sum_{\substack{A\subseteq J \\ |A|\ge 2}} (-1)^{|A|} T_A. \tag{97}$$

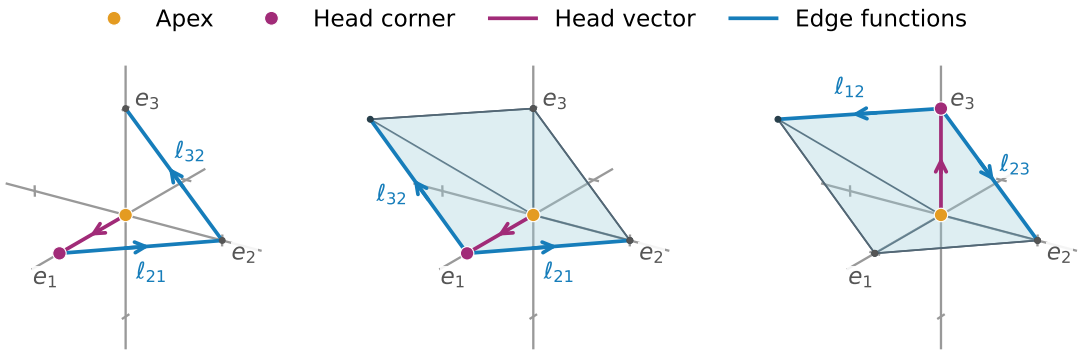
The empty term is a sum of support functions of segments. For $|A| \ge 2$, the polytope with support function $T_A$ has one head direction and $q - |A|$ segment directions, hence dimension at most $1 + q - |A| \le q - 1 = N - 3$.

Thus the singleton sum lies in $\mathcal{L}_{N-3}$; the sum of the linear apex coordinates lies in $\mathcal{L}_0$. Each fixed-position exchange is a bijection on lower label orders, so

$$q \sum_{\bar{\pi}} C_{\bar{\pi},\varnothing} = \sum_{\bar{\pi}} \sum_{a \in J} C_{\bar{\pi}^{(a)},\varnothing} \in \mathcal{L}_{q-1}. \tag{98}$$

Since $q > 0$, the full empty-cut sum belongs to $\mathcal{L}_{q-1} = \mathcal{L}_{N-3}$. The empty-cut coefficient $(-1)^{N-1}/(N-1)!$ and the factor $2/N$ from the two endpoint fibres are scalars, so the extra star sum lies there as well. This proves Claim 14.5. □

# B Network construction and width bounds




(a) Coordinate differences (b) Original description (c) Normalized description

Figure 3: Coordinate differences and the caterpillar polytope for $\pi = (0, 3, 2, 1)$ and $S = \{1, 2\}$. (a) The head $h = e_1 - e_0$ and edge functions $L = \{\ell_{32} = e_3 - e_2, \ell_{21} = e_2 - e_1\}$ drawn between their labelled endpoints. (b) The same generators translated to the base corner $h$ of $Q_{h,L}$. (c) Normalization (Section 6) gives $h^+ = h + \ell_{32} + \ell_{21} = e_3 - e_0$ and $L^+ = \{\ell_{12} = e_1 - e_2, \ell_{23} = e_2 - e_3\}$, leaving the polytope unchanged. Purple arrows represent the head (the apex–last spine connection before normalization); blue arrows represent the spine edge functions. Orange dots mark the polytope apex and purple dots the chosen head corner. The grey coordinate axes meet at the origin, representing $e_0$ after restriction to $x_0 = 0$; the positive unit points are labelled $e_1, e_2, e_3$.

**Proof of Theorem 1.** Fix $N \geq 3$. We bound the quotient dimension $q_N$ of Section 7 by the count of normalized descriptions and apply Theorem 15.

**Step 1.1 (Count the caterpillar face functions).** Use the adjusted heads and normalized descriptions from Section 6. Counting distinct normalized descriptions bounds $q_N$.

For a caterpillar description $(\pi, S)$, the spine endpoints are labelled $\pi(1)$ and $\pi(q+1)$, and every label other than $\pi(1)$ and the apex is the child of exactly one edge. Hence the normalized description of $U_{\pi,S}$ is

$$\begin{aligned} h^+_{\pi,S}(x) &= \sum_{\ell_{ij} \in L_{\pi,S}} x_{\max(i,j)} + x_{\pi(1)} + x_{\pi(q+1)} - \sum_{i=0}^{N-1} x_i, \\ L^+_{\pi,S} &= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in L_{\pi,S}\}. \end{aligned} \tag{99}$$

This depends only on the undirected tree, apex, and unordered spine endpoints. For $m$ spine vertices, choose the vertex set in $\binom{N}{m}$ ways, its order up to reversal in $m!/2$ ways, the apex in $N-m$ ways, and each of the $N-m-1$ leaf attachments in $m$ ways. Reversal has no fixed point since $m \geq 2$, so Equation (6) gives

$$\left|\{(h^+_{\pi,S}, L^+_{\pi,S}) : (\pi, S) \text{ a caterpillar description}\}\right| \leq C_N. \tag{100}$$

Reversing an edge swaps its omitted and linear states. Thus choosing states before or after normalization gives the same family $\mathcal{K}_0$ of descriptions, with $|\mathcal{K}_0| \leq 3^{N-2}C_N$. Write $U_k := U_{h,M}$ for $k = (h, M)$. We merge equal descriptions; distinct descriptions may still represent the same function.

Among them are the descriptions $(\ell_{ij}, \varnothing)$ representing $x \mapsto \sigma(x_i - x_j)$, with $i \neq j$: for the all-cut path $S = \{1, \dots, r\}$ with apex $\pi(0) = j$ and last spine vertex $\pi(r+1) = i$, omitting every edge function gives

$$U^0_{\pi,S} = \sigma \circ \ell_{ij} \qquad (0 \leq i \neq j < N). \tag{101}$$

Let $\mathcal{P} \subseteq \mathcal{K}_0$ consist of descriptions with empty hinge set and head a coordinate difference. The distinct heads $\ell_{ij}$ give $|\mathcal{P}| \geq N(N-1)$. Put

$$K := \mathcal{K}_0 \setminus \mathcal{P}. \tag{102}$$

Every face function in $\mathcal{D}$ equals $U_k$ for its normalized description $k \in \mathcal{K}_0$, and by Equation (8) every unit indexed by $\mathcal{P}$ lies in the auxiliary span $A_N$. Hence the units $U_k$, $k \in K$, span $V_N$ modulo $A_N$, and

$$q_N \leq |K| \leq 3^{N-2}C_N - N(N-1). \tag{103}$$

**Step 1.2 (Compile the network and establish its widths).** Theorem 15 gives a network with rational weights, zero biases, first width $h_1$, and second width

$$q_N + 2 \leq 3^{N-2}C_N - N(N-1) + 2 = h_2. \tag{104}$$

Zero padding realizes the stated widths exactly.

For the growth rates, count the ordered choices of $m$ spine labels:

$$\binom{N}{m}\frac{m!}{2} = \frac{N!}{2(N-m)!} \leq N^m \qquad (2 \leq m \leq N-1). \tag{105}$$

Using $N - m \leq N$ and $m^{N-m-1} \leq N^{N-m-1}$ in each summand of Equation (6) gives

$$\begin{aligned} C_N &\leq \sum_{m=2}^{N-1} N^m \, N \, N^{N-m-1} = (N-2)N^N \\ &\leq N^{N+1} \leq N^{4N}. \end{aligned} \tag{106}$$

Since $3^{N-2} \leq N^N$ for $N \geq 3$,

$$h_2 \leq 3^{N-2}(N-2)N^N + 2 \leq 2N^{5N}, \tag{107}$$

$$h_1 \leq 4N^2. \tag{108}$$

Taking logarithms gives the asymptotic bounds in Theorem 1. □

# C Proofs of the lower bounds

## C.1 Proof of the exponential count bound

**Lemma 16** (A common test at a tied maximum)**.** *Let $T$ have $m \geq 3$ labels, let $p$ have value zero on $T$ and distinct negative coordinates outside $T$, and let $\iota_T$ extend vectors by zero outside $T$. Write $\mathbf{1}_T$ for the all-ones vector in $\mathbb{R}^T$. Fix $t_0 \in T$. For a CPWL function $f$, define the linear test*

$$\mathcal{J}_{p,T} f := \partial_{\mathbf{1}_T}\big(y \mapsto \partial_p f(\iota_T y)\big)\big|_{y_{t_0}=0} \pmod{\mathcal{L}_{m-2}}. \tag{109}$$

*This test annihilates every shallow function and is nonzero on $\max_N$. For a unit $U = \sigma(A)$ of the form Equation* (35)*, a nonzero test requires $A(p) = 0$. Write its resulting tied germ as $g(y) = \sigma(H(y) + \sum_{i,j\in T} c_{ij}\sigma(y_i - y_j))$. Then it also requires $H(\mathbf{1}_T) = 0$ and at most two connected components in the comparison graph induced on $T$. If a tied label is isolated in that graph and its coefficient in $H$ is zero, the test vanishes.*

*Proof.* Directional germs and linear restriction preserve shallow functions. Both germs of the maximum in Equation (109) are $\max_{i\in T} y_i$. After fixing $y_{t_0} = 0$, this is the support function of an $(m-1)$-simplex and is not in $\mathcal{L}_{m-2}$. Otherwise successive opposite-face differences and facet restrictions, using Equation (46), would put a segment support function in $\mathcal{L}_0$, although that function is not linear.

If $A(p) \neq 0$, the first germ of $U$ is zero or shallow. Otherwise it is $g$ as stated. Put $\lambda = H(\mathbf{1}_T)$. At $\mathbf{1}_T$, all internal comparisons vanish, so the second germ is zero when $\lambda < 0$, shallow when $\lambda > 0$, and exactly $g$ when $\lambda = 0$. Only the last case can survive, and then $g$ is invariant under common translation.

If the induced graph has $c$ components, its comparison forms span $m - c$ directions. Adjoining $H$ gives at most $m - c + 1$ directions. This bounds the polytope dimension even for signed coefficients: split the preactivation as $h_P - h_Q$ in that span and use $\sigma(h_P - h_Q) = h_{\mathrm{conv}(P\cup Q)} - h_Q$. Thus $c \geq 3$ makes the test zero. If an isolated label has zero coefficient in $H$, all forms annihilate both its coordinate vector and $\mathbf{1}_T$. These independent vectors leave at most $m - 2$ directions, so this case also vanishes, including when the isolated label is $t_0$. □

Consequently, in every exact expansion the units with nonzero test cover all top ties. We now bound their coverage in two ways: coefficient balance for oriented forest functions, and induced components for weighted graphs. Their capacities can be added because they use the same test.

We prove Theorem 9 and obtain Equations (31) to (32) by taking $\Delta = 2$. We use Lemma 16 and a product over disjoint star blocks.

**Proof of Theorem 8.** Fix an expansion as in Theorem 9, and $3 \leq m \leq N$. Write $U_k = \sigma(A_k)$ with $A_k(x) = \sum_o h_{k,o}x_o + \sum_{(i,j)\in M_k} \sigma(x_i - x_j)$. Reorienting edges by Equation (8) preserves $A_k$ and $\kappa_{k,o}$, so we may assume $i < j$ in $M_k$. Choose an integer $B > 1$ with $B > \sum_o |h_{k,o}| + |M_k|$ for every $k$. A permutation $\rho$ assigns positions to labels, with smaller positions higher. Put $T_\rho = \{o : \rho(o) < m\}$ and

$$p^\rho_o = \begin{cases} 0, & o \in T_\rho, \\ -B^{\rho(o)}, & o \notin T_\rho. \end{cases} \tag{110}$$

By Lemma 16, some unit has nonzero test at this locality.

**Step 8.1 (Detection forces balance).** Fix such a unit and suppress its index. The common test forces $A(p^\rho) = 0$. For $o \notin T_\rho$, let $u_\rho(o)$ count its neighbours at smaller positions. Collecting coefficients gives

$$A(p^\rho) = -\sum_{o\notin T_\rho} (\kappa_o - u_\rho(o))B^{\rho(o)}. \tag{111}$$

The integer digits have absolute value less than $B$. Successively reducing modulo $B$ forces each digit to vanish. If a tied label $t$ has no tied neighbour, its coefficient in the tied head is $\kappa_t$. The last assertion of Lemma 16 requires this coefficient to be nonzero. Call $U$ *passing* $\rho$ when these two necessary conditions hold:

$$u_\rho(o) = \kappa_o \qquad \text{for } o \notin T_\rho, \tag{112}$$

$$\kappa_t \neq 0 \qquad \text{for } t \in T_\rho \text{ with no neighbour in } T_\rho. \tag{113}$$

Every permutation is passed by some unit in the expansion.

**Step 8.2 (Only the reduced forest chooses the tie set).** Fix a unit, let $F$ be its forest, and call the deleted degree-one vertices *pendants*. Let $\Phi$ contain the vertices that are neither pendants nor isolated in $F$. For a passed tie set $T$, an isolated vertex belongs to $T$ exactly when its $\kappa$ is nonzero. A pendant belongs to $T$ exactly when its neighbour does: either mismatch contradicts Equations (112) and (113). A component consisting of two pendants must lie entirely in $T$, since otherwise its lower vertex violates balance. Consequently $T$ is determined by $A = T \cap \Phi$, and $|A| \leq m$.

**Step 8.3 (Each disjoint edge halves the possible orders).** For fixed $T$, balance prescribes the out-degree of each vertex of $F - T$ when edges point towards the higher endpoint. A forest orientation is uniquely determined by its out-degrees: orient the edge at a leaf, delete the leaf, and repeat. Hence every passed permutation with this $T$ induces the same orientation of $F - T$. If we select a matching of $\mu(A)$ edges of $F - T$, independently swapping their endpoints gives

$$|\{\rho : U \text{ passes } \rho,\ T_\rho = T\}| \leq m!\,(N-m)!\,2^{-\mu(A)}. \tag{114}$$

**Step 8.4 (Partition into blocks and multiply their weights).** Partition $\Phi$ into stars of the reduced forest and singletons isolated in that forest. Such a star partition exists: an inclusion-minimal edge cover of the non-isolated vertices contains no three-edge path, since its middle edge could be removed. As it is a forest, its components are stars. Every star has $1 \leq s \leq \Delta$ leaves. A singleton has a pendant neighbour in $F$; fix one.

For $A \subseteq \Phi$, select one edge in a star block whenever its centre and some leaf are outside $A$. At a singleton outside $A$, select its fixed pendant edge. For a passed tie set these are disjoint edges of $F - T$. Let $\mu(A)$ count them and put $y = 1/(2\Delta + 2) = 1/b$. The sum of $y^{|A|}2^{-\mu(A)}$ factorizes over blocks. A singleton has weight $1/2 + y$; a star with $s$ leaves has weight

$$W_s(y) = y(1+y)^s + \tfrac{1}{2}\big((1+y)^s - y^s\big) + y^s = (\tfrac{1}{2} + y)(1+y)^s + \tfrac{1}{2}y^s. \tag{115}$$

The three terms distinguish a selected centre, an unselected centre with an unselected leaf, and an unselected centre with all leaves selected. All block weights are at most 1. For $\Delta = 2$, the singleton, one-leaf, and two-leaf weights at $y = 1/6$ are respectively $2/3$, $31/36$, and $199/216$. In general $y \leq 1/6$; the one-leaf case is bounded by $31/36$, and for $2 \leq s \leq \Delta$, $W_s(y) \leq (1/2 + y)e^{1/2-y} + y^2/2 \leq (2/3)e^{1/3} + 1/72 < 1$. Thus

$$\sum_{A \subseteq \Phi} y^{|A|}2^{-\mu(A)} = \prod_{\text{blocks}} W_{\text{block}}(y) \leq 1. \tag{116}$$

**Step 8.5 (Count all permutations).** Distinct passed tie sets have distinct $A = T \cap \Phi$, and $|A| \leq m$. Summing Equation (114) and inserting the factor $1 \leq b^m y^{|A|}$ bounds the number of permutations passed by one unit by $m!\,(N-m)!\,b^m$ using Equation (116). The units cover all $N!$ permutations, so $N! \leq |K|\,m!\,(N-m)!\,b^m$, proving Equation (34). Caterpillar functions have reduced degree at most 2, giving Equation (31).

**Step 8.6 (Extract the exponential bound).** Put $t_m = \binom{N}{m} b^{-m}$. If $N \geq 3b + 2$, then $t_0 \leq t_1 \leq t_2 \leq t_3$, so every $t_m$ is at most $|K|$. The binomial theorem gives

$$(1 + 1/b)^N = \sum_{m=0}^{N} t_m \leq (N + 1)|K|. \tag{117}$$

For $b = 6$ and $3 \leq N < 20$, the same conclusion follows from $|K| \geq 1$ and $(7/6)^N \leq N + 1$. This proves Equation (32) and the remaining assertions of Theorem 9.

**Quadratic refinement (`Lean_29`).** In fact the same bound holds for every canonical tree face function, as stated in Corollary 12. Use the common test on triples. Coefficient balance forces the original apex into the tied set, since its coefficient is $-1$ and no edge meets it. The apex is therefore isolated in the induced comparison graph. By Lemma 16, the other two tied labels must be the endpoints of a retained edge. Each unit has at most $N - 2$ such edges, so covering all triples gives $\binom{N}{3} \leq (N - 2)|K|$, or $N(N - 1) \leq 6|K|$. This proves both Equation (33) and corollary 12. □

**Proof of Theorem 10.** First remove biases, if present, by taking the recession function $\lim_{t\to\infty} F(tx)/t$. This preserves the homogeneous target, the comparison graphs, and the number of neurons.

By Lemma 16, each $m$-element tie needs a unit whose induced comparison graph has at most two connected components.

For a graph of maximum degree two, let $C_j$ count its connected induced $j$-vertex subsets. Each path or cycle component contributes at most its number of vertices: a proper connected subset is an interval or circular arc, and a whole component contributes only once. Hence $C_j \leq N$. An $m$-element set with two induced components is an unordered union of two connected sets. Dropping disjointness and the absence of edges between them only increases the count. A single unit consequently covers at most

$$C_m + \frac{1}{2} \sum_{j=1}^{m-1} C_j C_{m-j} \leq N + \frac{m - 1}{2} N^2 \tag{118}$$

ties of size $m$. Counting all $\binom{N}{m}$ ties proves Equation (36). The central binomial coefficient is $\Theta(2^N/\sqrt{N})$, giving the stated asymptotic bound. □

**Proof of Theorem 11.** By Lemma 16, every label set $T$ with $|T| \geq 3$ needs a comparison graph $G$ for which $G[T]$ has at most two components. Give a pseudoforest $G$ an auxiliary orientation of indegree at most one: orient tree components away from a root, cycles cyclically, and attached trees away from the cycle. This orientation is independent of the ReLU orientations. Every connected induced subgraph has at most one vertex of indegree zero, since their number is $|V| - |E|$. Fix a matching. If a matched parent is absent from $T$ and its child is present, that child has indegree zero in $G[T]$. Different matching edges therefore give different components. With independent vertex selection of probability $p$, these events are independent and have probability $p(1 - p)$. Thus, for a matching of size $k$ and $\rho = 1 - p(1 - p)$,

$$\Pr\big(c(G[T]) \leq 2\big) \leq \rho^k \left(1 + k\frac{1 - \rho}{\rho} + \binom{k}{2}\left(\frac{1 - \rho}{\rho}\right)^2\right). \tag{119}$$

At $p = 1/2$, selected isolated vertices give additional independent component witnesses of probability $1/2$, on labels disjoint from the matching. Replacing their probability by $1/4$ only increases the probability

of at most two witnesses. A pseudoforest with $\nu(G) + z(G) \geq k$ therefore serves at most a fraction $(3/4)^k c_k$ of all subsets. Since $\Pr(|T| \geq 3) = 1 - b_N 2^{-N}$, the union bound proves Equation (37).

For the degree bound, fix neighbor orders in an arbitrary graph $G$ of maximum degree $\Delta \geq 3$. A connected vertex set containing a specified root has a deterministic breadth-first spanning tree. Encode each child by its neighbor slot, omitting the parent slot at nonroot vertices. The root has at most $\Delta$ slots and the other vertices at most $d = \Delta - 1$. Following the slots recovers the original set, so this encoding is injective. The planted-tree and root generating series satisfy $B = z(1 + B)^d$ and $R = z(1 + B)^{d+1} = B + B^2$. Lagrange inversion and Stirling give

$$r_m := [z^m]R = \frac{1}{m}\left(\binom{dm}{m-1} + 2\binom{dm}{m-2}\right) = O_\Delta(\alpha^m m^{-3/2}), \qquad \alpha = 1/y_\Delta.$$

If $C_m$ counts connected $m$-vertex sets, rooting each set at all its vertices gives $C_m \leq N r_m/m$. The number $P_m$ of sets with at most two induced components therefore satisfies

$$P_m \leq C_m + \frac{1}{2}\sum_{j=1}^{m-1} C_j C_{m-j} = O_\Delta(N^2 \alpha^m m^{-5/2}). \tag{120}$$

Here the convolution has the same power $m^{-5/2}$ because $\sum_{j\geq 1} j^{-5/2}$ converges. Take $m$ nearest $N/(1+\alpha)$. Then $P_m = O_\Delta(N^{-1/2}\alpha^m)$, while $\binom{N}{m}\alpha^{-m} = \Theta_\Delta((1 + y_\Delta)^N/\sqrt{N})$. Since $\binom{N}{m} \leq |K|P_m$, this proves Equation (38). Repeating a graph with different heads and weights adds no tied sets to its coverage, so these bounds also count distinct graphs. □

For Equation (39), extend a maximum matching of $G$ to a spanning forest, then add one omitted edge in each originally cyclic component. The resulting pseudoforest $P$ preserves the matching and all isolated vertices, and omits exactly $b(G)$ edges. Restoring an edge merges at most two induced components, so $c(G[T]) \geq c(P[T]) - b(G)$. The independent matching and isolate witnesses in the preceding proof therefore show that $c(G[T]) \leq 2$ requires at most $b + 2$ successes in a binomial variable with parameters $k$ and $1/4$. Apply the same union bound. The exponential conclusions follow from the binomial lower-tail estimate; when $b = o(k)$ its probability is at most $\exp[-(\log(4/3) - o(1))k]$.

For mixtures with the original caterpillar and rooted-star face dictionaries, their normalized size-$m$ capacity is at most $6^m$. Thus the total number $Z$ of functions satisfies $\binom{N}{m} \leq Z(6^m + O_\Delta(N^2\alpha^m m^{-5/2}))$. For added graphs of degree at most three, use $\alpha = 4$ and $m$ nearest $N/7$; the $6^m$ term dominates and gives $Z = \Omega((7/6)^N/\sqrt{N})$. For $\Delta \geq 4$, we have $\alpha \geq 27/4 > 6$; take $m$ nearest $N/(1+\alpha)$, and the connected-set term gives $Z = \Omega_\Delta((1 + y_\Delta)^N)$ as above.

**Arbitrary tree shapes with a source orientation.** The block argument also works with a subforest $R$ of the reduced forest: its maximum degree must be at most two, and each vertex of $\Phi$ isolated in $R$ must either have a pendant neighbour or belong to every passed tie set. Use the same star partition of $R$. The first kind of singleton has weight $1/2 + y$; the second has weight $y$, since it must be selected. All weights are at most one at $y = 1/6$, so the same counting argument applies. A leaf with $\kappa \notin \{0, 1\}$ is always tied: otherwise balance would require its unique edge to give an out-degree other than zero or one.

For $U_{a,s,r,T}$, root the underlying tree at the head root $r$. Call a nonleaf *poor* if it has neither a pendant nor an out-leaf neighbour. Choose one child of each poor vertex; such a child exists, and no child is chosen by two parents. Let $R$ contain these chosen edges, together with every edge from an out-leaf to a nonpendant neighbour. These edges belong to the reduced forest. An isolated vertex of $R$ in $\Phi$ is either adjacent to a pendant or a leaf with $\kappa \notin \{0, 1\}$, so it satisfies the required condition.

Leaves have degree at most one in $R$. A nonleaf with at most one out-leaf neighbour has degree at most two in $R$: a poor vertex contributes one chosen child edge and can receive one, whereas any other

vertex receives at most one chosen child edge. The only remaining case is a common neighbour $v$ of the two out-leaves $s$ and $r$. It is not poor. Its parent is the leaf $r$, which is not poor either, so no chosen child edge ends at $v$. Its only $R$-edges are the two out-leaf edges. Thus $R$ has maximum degree two. This gives the per-unit factor $6^m$ for every $U_{a,s,r,T}$, and hence the stated $(7/6)^N/(N+1)$ bound for mixtures with the existing caterpillar and rooted-star face dictionaries.

**Combining coefficient balance and large matchings.** Let $K_0$ contain the functions whose ordered-locality bound is $6^m$: reduced-degree-two forest functions and the source-oriented functions above. Let $K_1$ contain the added weighted-pseudoforest functions, each with a matching of size at least $k$, and put $Z = |K_0| + |K_1|$. Both necessary conditions come from the same test in Lemma 16; its nonzero simplex class must be detected by a term in one of the two families.

Take $m = \lfloor N/4 \rfloor$, $p = 1/4$, and $\rho = 13/16$. Give each size-$m$ tie the weight $p^m(1-p)^{N-m}$ and average uniformly over the orders of its outside coordinates. The total target weight is $P_m = \binom{N}{m} p^m (1-p)^{N-m}$. By Equation (119), each added function covers weight at most $\rho^k d_k$. Each base function covers at most $6^m p^m (1-p)^{N-m} = (3/4)^N 2^m \le \rho^{N/2} \le \rho^k$; the first inequality follows from $162 \le 169$ by taking fourth powers of the exponential bases. Since $d_k \ge 1$, counting the union gives $P_m \le Z\rho^k d_k$. The chosen $m$ is a mode of $\mathrm{Bin}(N, 1/4)$, so $P_m \ge 1/(N+1)$. This proves Equation (40).

For added graphs of excess at most $b$, replace $d_k$ by $\sum_{j=0}^{\min(k,b+2)} \binom{k}{j} (3/13)^j$: the pseudoforest reduction above permits at most $b+2$ matching witnesses. If $k = \Omega(N)$ and $b = o(N)$, this sum is $\exp(o(N))$, and the bound becomes $Z \ge \exp[k \log(16/13) - o(N)]$. Finally, for a tree on $n$ vertices, the leaf criterion follows from a minimum vertex cover $C$ of size equal to the maximum matching number $\nu$. The nonleaf vertices outside $C$ form an independent set $I$; their at least $2|I|$ incident edges lie in the forest on $I \cup C$, which has at most $|I| + \nu - 1$ edges. Hence $|I| \le \nu - 1$, and $n - L \le 2\nu - 1$.

**Adding every oriented star.** Work in full coordinates. For a label $j$, let $\iota_j$ insert a zero at that coordinate and put $\mathcal{T}_j g = (\partial_{-e_j} g - \partial_{e_j} g) \circ \iota_j$. This linear operator sends $\mathcal{L}_d$ into $\mathcal{L}_{d-1}$ for $d \ge 1$ and annihilates $\mathcal{L}_0$: if coordinate $j$ is constant on a polytope, its two exposed faces coincide; otherwise both have smaller dimension. Projection under restriction cannot increase dimension. Moreover, $\mathcal{T}_j \max_m = \max_{m-1}$, since the simplex faces are its facet opposite $e_j$ and the vertex $e_j$.

For a canonical star, deletion of a nonroot leaf gives the smaller star modulo $\mathcal{L}_1$, with sign $+$ for $\sigma(x_l - x_c)$ and sign $-$ for $\sigma(x_c - x_l)$. If the deleted leaf is the head root, the latter orientation transfers the head to the centre with sign $+$; the former gives zero modulo $\mathcal{L}_1$. Deleting the apex leaves a zonotope, and deleting the centre fixes every segment, so these cases also give zero modulo $\mathcal{L}_1$. Thus successive deletions preserve the span of canonical stars modulo the lower-dimensional terms.

At five labels, a six-flag certificate gives a linear functional that annihilates every canonical oriented star and $\mathcal{L}_3$, and takes value $120$ on the simplex. The independent vertex checker checks all $640$ labelled stars, with every head root and orientation, by successive maxima and minima over explicit vertex sets. Composing this functional with $m-5$ coordinate deletions therefore gives such a separator at every $m \ge 5$. A nonhinge state already has dimension at most $m-2$, so all star face states are covered too.

Assign every term whose function belongs to the star family to the added star class, even if it was originally listed in another class. Here $K_0, K_1$ count only the remaining terms, and $Z = |K_0| + |K_1|$. At any tie of size at least five, a surviving top-dimensional star germ is again a canonical star: each untied component has a unique attachment to the tied subtree, and summing its zero frozen coefficients transports the head to that attachment if it contains the original root. Apply the separator to the local expansion. Every added star term vanishes, so a term from the other classes must cover the locality. With caterpillars alone, count at $m$ nearest $N/7$ to obtain $\binom{N}{m} \le 6^m |K_0|$ and hence $|K_0| = \Omega((7/6)^N/\sqrt{N})$. The bounded-degree mixture proofs use tied sizes growing linearly with $N$, so they too apply unchanged once $m \ge 5$. For the matching

class, $m = \lfloor N/4 \rfloor \geq 5$ when $N \geq 20$; the proof of Equation (40) therefore gives the same bound without counting the added star terms.

### C.2 Proof of the dimension bound

**Proof of Theorem 13.** Put $r = N - 2$ and $b = N - 1$. We exhibit $r!$ dictionary functions and linear functionals whose evaluation matrix is the identity and which annihilate $A_N$.

**Step 13.1 (One path for each order).** For every permutation $\pi$ fixing 0 and $b$, use the path $\pi(1), \ldots, \pi(r), b$ with apex 0 and keep every edge as a hinge:

$$U_\pi(x) = \sigma\Big(x_b - x_0 + \sum_{j=1}^{r} \sigma(x_{\pi(j)} - x_{\pi(j+1)})\Big) \in V_N. \tag{121}$$

There are $r!$ such paths.

**Step 13.2 (One mixed difference).** For an order $\rho$, set $c = \rho(r)$ and

$$z^\rho_0 = z^\rho_b = 2r, \qquad z^\rho_{\rho(j)} = 2j \quad (1 \leq j \leq r). \tag{122}$$

Thus $0, b, c$ tie at the top. Let $\Lambda_\rho g = \sum_{S \subseteq \{0,b,c\}} (-1)^{3-|S|} g(z^\rho + \mathbf{1}_S)$. This third mixed difference annihilates every function of at most two coordinates, by pairing corners along a missing coordinate. In particular, $\Lambda_\rho A_N = 0$.

**Step 13.3 (The evaluation matrix is the identity).** On the cube of $\Lambda_\rho$, every inner function of $U_\rho$ except the last is zero: adjacent base heights differ by 2, while a coordinate moves by at most 1. Writing $v = x - z^\rho$, we obtain $U_\rho = \sigma(v_b - v_0 + \sigma(v_c - v_b))$. At the eight corners this is 1 exactly when $v_0 = 0$ and $(v_b, v_c) \neq (0, 0)$, so its mixed difference is $1 + 1 - 1 = 1$.

If $\pi \neq \rho$, the middle labels in the $\pi$-order have a descent in $\rho$-height. The corresponding inner function is at least 1 everywhere on the cube, whereas $x_b - x_0 \geq -1$. The outer ReLU can therefore be removed there. The resulting sum depends on at most two coordinates term by term, so its third mixed difference is zero. Hence

$$\Lambda_\rho U_\pi = \begin{cases} 1, & \pi = \rho, \\ 0, & \pi \neq \rho. \end{cases} \tag{123}$$

**Step 13.4 (Independence modulo the auxiliary span).** Applying $\Lambda_\rho$ to $\sum_\pi c_\pi U_\pi \in A_N$ gives $c_\rho = 0$. Thus these $r!$ functions are independent modulo $A_N$, proving $q_N \geq (N-2)!$. □

## D Relation to the construction of Brunck et al.

The combinatorial proof of [BHR+26, Part 2] cones zonotopes $B_{v,\pi,M}$ to pyramids and expresses the simplex support function as a signed sum of their support functions modulo lower-dimensional terms [BHR+26, Lemma 22]. This appendix identifies these building blocks. For a nonempty set $M$ of marked positions, the pyramid has the support function of a translated caterpillar polytope, with apex $v$, label order $\pi$, and cut set $M - 1$. For $M = \varnothing$ it gives a rooted star, which is not a caterpillar function. We use the indexing of [BHR+26, Section 7.1], with their $n$ written as $N$ and $[N]$ identified with our label set $\{0, \ldots, N-1\}$. Positions of an ordering are numbered from 1.

**Definition 17** (Zonotopes of Brunck et al., Lean_30, Lean_31)**.** Let $N \geq 3$, let $v$ be a label, let $\pi = (\pi_1, \dots, \pi_{N-1})$ be an ordering of the remaining labels, and let $M \subseteq \{2, \dots, N-1\}$. For $p \in \mathbb{R}^N$ put $u_i(p) := \sum_{1 \leq j < i} p_{\pi_j}$. Following [BHR+26, Equation (3)], define

$$H_{-v} := \Big\{ p \in \mathbb{R}^N : p_v = 0,\ \sum_k p_k = 1 \Big\}, \tag{124}$$

$$B_{v,\pi,M} := \left\{ p \in H_{-v} : \begin{array}{ll} 0 \leq u_i(p) \leq 1 & \text{for } i \in M, \\ -1 \leq p_{\pi_i} \leq 0 & \text{for } i \in \{2, \dots, N-1\} \setminus M \end{array} \right\}. \tag{125}$$

The pyramid with apex $e_v$ is $\mathrm{pyr}_{e_v}(B_{v,\pi,M}) := \mathrm{conv}(\{e_v\} \cup B_{v,\pi,M})$, with support function $h_{v,\pi,M}$ as in Equation (16). Its last marked position and the marked parent of a position $i$ are

$$q := \max(\{1\} \cup M), \qquad r(i) := \max\big(\{1\} \cup \{j \in M : j < i\}\big) \qquad (2 \leq i \leq N-1). \tag{126}$$

For $M \neq \varnothing$, $q = \max M$. For labels $a \neq c$, the *rooted star* with apex $a$ and root $c$ is

$$S_{a,c}(x) := \sigma\Big( x_c - x_a + \sum_{l \notin \{a,c\}} \sigma(x_c - x_l) \Big). \tag{127}$$

**Theorem 18** (Relation to Brunck et al., Lean_32, Lean_33, Lean_34, Lean_35, Lean_36, Lean_37, Lean_38, Lean_39, Proof)**.** *Let $v$, $\pi$, and $M$ be as in Definition 17, and let $q$ and $r(i)$ be as in Equation* (126)*.*

*(a) The base is the zonotope*

$$B_{v,\pi,M} = e_{\pi_q} + \sum_{i=2}^{N-1} \big[0, e_{\pi_{r(i)}} - e_{\pi_i}\big]. \tag{128}$$

*(b) For every $x \in \mathbb{R}^N$,*

$$h_{v,\pi,M}(x) = x_v + \sigma\Big( x_{\pi_q} - x_v + \sum_{i=2}^{N-1} \sigma\big(x_{\pi_{r(i)}} - x_{\pi_i}\big) \Big). \tag{129}$$

*(c) Let $M \neq \varnothing$. Define the caterpillar description $(\pi', S)$ by $\pi'(0) := v$, $\pi'(j) := \pi_j$ for $1 \leq j \leq N-1$, and $S := \{i - 1 : i \in M\}$. Then $|S| = |M|$ and*

$$h_{v,\pi,M}(x) = x_v + U_{\pi',S}(x) \qquad (x \in \mathbb{R}^N). \tag{130}$$

*Hence $h_{v,\pi,M} - x_v = U_k$ for some $k \in \mathcal{K}_0$. Conversely, every caterpillar description $(\pi', S)$ arises in this way from $v = \pi'(0)$, $\pi_j = \pi'(j)$, and $M = \{j + 1 : j \in S\} \neq \varnothing$.*

*(d) Let $M = \varnothing$. Then $h_{v,\pi,\varnothing}(x) = x_v + S_{v,\pi_1}(x)$. No rooted star is a caterpillar function: $S_{a,c} \neq U_k$ for all labels $a \neq c$ and all $k \in \mathcal{K}_0$. In particular, $h_{v,\pi,\varnothing} - x_v \neq U_k$ for every $k \in \mathcal{K}_0$.*

By Equation (17), the right-hand side of Equation (130) is the support function of the translated caterpillar polytope $e_v + Q_{h_{\pi',S}, L_{\pi',S}}$. The family of [BHR+26] therefore consists of the caterpillar polytopes, translated by $e_v$, and of the pyramids over rooted stars. The sign $(-1)^{|M|}$ of [BHR+26, Lemma 22] equals $(-1)^{|S|}$. The theorem compares building blocks only. It makes no statement about corrections or network coefficients.

**Proof of Theorem 18.** Write $I := \{2, \dots, N-1\}$ and $g_i := e_{\pi_{r(i)}} - e_{\pi_i}$ for $i \in I$. Every element of $\{1\} \cup M$ lies in $\{1, \dots, N-1\}$, so $q$ and all $r(i)$ are positions. Since $M \subseteq I$, the definitions give

- $1 \le r(i) < i$ for $i \in I$, and $i \le r(l)$ whenever $i \in M$ and $i < l$,
- $i \le q$ for $i \in M$,
- $q \ne i$ and $r(l) \ne i$ for $i \in I \setminus M$ and all $l$.

**Step 18.1 (Coordinates of the zonotope).** For $t \in [0,1]^I$ let $z(t) := e_{\pi_q} + \sum_{l \in I} t_l g_l$. Every $e_{\pi_j}$ has coordinate sum 1 and vanishes at $v$, so $z(t) \in H_{-v}$. For $i \in I \setminus M$ the coordinate $z(t)_{\pi_i}$ collects the term $-t_i$ and no other term, since $\pi_i$ is neither $\pi_q$ nor any $\pi_{r(l)}$. Hence $z(t)_{\pi_i} = -t_i$. For $i \in M$, the prefix sum $u_i$ counts a vector $e_{\pi_j}$ exactly when $j < i$. Thus $e_{\pi_q}$ contributes 0, because $q \ge i$. A generator $g_l$ with $l < i$ contributes $1 - 1 = 0$, because $r(l) < l < i$. A generator with $l > i$ contributes $0 - 0$, because $r(l) \ge i$. Finally $g_i$ contributes $t_i$, because $r(i) < i$. Hence $u_i(z(t)) = t_i$. Therefore $z(t) \in B_{v,\pi,M}$.

**Step 18.2 (The coordinates determine the point).** Let $p, p' \in H_{-v}$ with $u_i(p) = u_i(p')$ for $i \in M$ and $p_{\pi_i} = p'_{\pi_i}$ for $i \in I \setminus M$. Put $d := p - p'$ and $U_i := \sum_{1 \le j < i} d_{\pi_j}$. Then $U_1 = 0$, and $U_N = \sum_k d_k - d_v = 0$. Descending from $i = N-1$ to $i = 2$, either $i \in M$ and $U_i = 0$ by assumption, or $i \notin M$ and $U_i = U_{i+1} - d_{\pi_i} = U_{i+1} = 0$. Hence $d_{\pi_j} = U_{j+1} - U_j = 0$ for $1 \le j \le N-1$, and $d_v = 0$. Thus $p = p'$. This is the invertibility in [BHR+26, Claim 21].

**Step 18.3 (Proof of (a)).** Step 18.1 shows that the right-hand side of Equation (128) lies in $B_{v,\pi,M}$. Conversely, let $p \in B_{v,\pi,M}$, and set $t_i := u_i(p)$ for $i \in M$ and $t_i := -p_{\pi_i}$ for $i \in I \setminus M$. The inequalities in Equation (125) say $t \in [0,1]^I$. By Step 18.1, the points $p$ and $z(t)$ of $H_{-v}$ have the same coordinates, so $p = z(t)$ by Step 18.2.

**Step 18.4 (Proof of (b)).** A linear function attains the same supremum on a set and on its convex hull. Hence $h_{v,\pi,M}(x) = \max(x_v, \sup_{p \in B_{v,\pi,M}} \langle p, x \rangle)$. Put $b_l := x_{\pi_{r(l)}} - x_{\pi_l}$. By (a), $\langle z(t), x \rangle = x_{\pi_q} + \sum_l t_l b_l$, and $t_l b_l \le \sigma(b_l)$ for $0 \le t_l \le 1$, with equality for $t_l = \mathbb{1}_{\{b_l \ge 0\}}$. So the supremum over $B_{v,\pi,M}$ equals $A := x_{\pi_q} + \sum_l \sigma(b_l)$. Then Equation (7) gives $\max(x_v, A) = x_v + \sigma(A - x_v)$, which is Equation (129).

**Step 18.5 (Proof of (c)).** With $\pi'$ and $S$ as in (c), position $j$ of the caterpillar carries the label $\pi'(j+1) = \pi_{j+1}$, so caterpillar position $j$ is the position $j+1$ of [BHR+26]. The map $i \mapsto i - 1$ is a bijection from $M$ onto $S$, hence $|S| = |M|$, and it carries $\{1\} \cup M$ onto $\{0\} \cup S$ and $\{1\} \cup \{j \in M : j < i\}$ onto $\{0\} \cup \{j \in S : j < i - 1\}$. Comparing with Equation (10), the last spine position is $q - 1$ and the parent of caterpillar position $i - 1$ is $r(i) - 1$. By Equations (12) and (13), the head is $x_{\pi_q} - x_v$ and the edge functions are $x_{\pi_{r(i)}} - x_{\pi_i}$, $i \in I$. So the right-hand side of Equation (129) is $x_v + U_{\pi',S}(x)$. If $M \ne \varnothing$, then $S \ne \varnothing$ and $0 \notin S$, so $(\pi', S)$ is a caterpillar description, and its normalized description $k \in \mathcal{K}_0$ satisfies $U_k = U_{\pi',S}$ by Equation (8). The converse inverts these bijections.

**Step 18.6 (Proof of (d)).** For $M = \varnothing$, $q = 1$ and $r \equiv 1$, and $\{\pi_i : i \in I\}$ is the set of labels other than $v$ and $\pi_1$. So Equation (129) becomes $x_v + S_{v,\pi_1}(x)$. Let $a \ne c$ and evaluate at $x = e_c$. Every inner term of $S_{a,c}(e_c)$ equals $\sigma(1)$, so $S_{a,c}(e_c) = 1 + (N-2) = N - 1$. Let $k \in \mathcal{K}_0$. Then $U_k$ is the normalization of a face function $U^{\tau}_{\pi',S}$, and $U_k = U^{\tau}_{\pi',S}$ by Equation (8). Since $\ell \le \sigma \circ \ell$, $0 \le \sigma \circ \ell$, and $\sigma$ is monotone, $U^{\tau}_{\pi',S} \le U_{\pi',S}$. At $x = e_c$, every head and edge function takes values in $\{-1, 0, 1\}$. If $c = \pi'(0)$, the argument of the outer $\sigma$ in $U_{\pi',S}(e_c)$ is $-1$, so $U_{\pi',S}(e_c) = 0$. Otherwise $c$ is the label of a caterpillar position $m$. The head contributes at most 1, and every inner hinge at most 1. If $m \ge 1$, the inner hinge of the edge into position $m$ is $\sigma(x_{\pi'(p(m)+1)} - 1) = 0$, so the value is at most $1 + (N-3)$. If $m = 0$, the head contributes 0, because $q \ge \min S > 0$, so the value is at most $N - 2$. Thus $U_k(e_c) \le N - 2 < S_{a,c}(e_c)$. □

# E Lean definitions and main theorem

The real parameter arrays in Equation (4) have dimensions

$$\begin{aligned} W_1 &\in \mathbb{R}^{h_1 \times N}, \quad b_1 \in \mathbb{R}^{h_1}, \\ W_2 &\in \mathbb{R}^{h_2 \times h_1}, \quad b_2 \in \mathbb{R}^{h_2}, \\ w_3 &\in \mathbb{R}^{h_2}, \qquad b_3 \in \mathbb{R}, \end{aligned} \tag{131}$$

**Sources (`Definitions.lean`, `Theorems.lean`).** The listings specify the network, widths, and correctness for every real input. They reproduce declarations from these files at revision `8b4d74ab66f4`, omitting imports, documentation comments, and the theorem's proof body; these remain in the linked sources.

## E.1 Network and width definitions

```
namespace MaxCaterpillar

open BigOperators

abbrev Point (N : ℕ) := Fin N → ℝ

noncomputable def relu (t : ℝ) : ℝ := max 0 t

structure Network (N h₁ h₂ : ℕ) where
  W₁ : Fin h₁ → Fin N → ℝ
  b₁ : Fin h₁ → ℝ
  W₂ : Fin h₂ → Fin h₁ → ℝ
  b₂ : Fin h₂ → ℝ
  w₃ : Fin h₂ → ℝ
  b₃ : ℝ

noncomputable def Network.eval {N h₁ h₂ : ℕ} (net : Network N h₁ h₂)
    (x : Point N) : ℝ :=
  (∑ k, net.w₃ k * relu
    ((∑ j, net.W₂ k j * relu
      ((∑ i, net.W₁ j i * x i) + net.b₁ j)) + net.b₂ k))
    + net.b₃

def Network.HasRationalWeights {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  (∀ j i, ∃ q : ℚ, (q : ℝ) = net.W₁ j i) ∧
    (∀ k j, ∃ q : ℚ, (q : ℝ) = net.W₂ k j) ∧
    ∀ k, ∃ q : ℚ, (q : ℝ) = net.w₃ k

def Network.HasZeroBiases {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  net.b₁ = 0 ∧ net.b₂ = 0 ∧ net.b₃ = 0
```

```
noncomputable def Maximum (N : ℕ) [NeZero N] (x : Point N) : ℝ :=
  Finset.univ.sup'
    ⟨⟨0, Nat.pos_of_ne_zero (NeZero.ne N)⟩,
      Finset.mem_univ _⟩ x

def firstWidth (N : ℕ) : ℕ := N.choose 2 + N + 1

def coneCount (N : ℕ) : ℕ :=
  ∑ m in Finset.Icc 2 (N - 1),
    N.choose m * (m.factorial / 2) * (N - m)
      * m ^ (N - m - 1)

def secondWidth (N : ℕ) : ℕ :=
  coneCount N * 3 ^ (N - 2) - N * (N - 1) + 2

end MaxCaterpillar
```

## E.2 Caterpillar functions and expansions

These definitions, also in `Definitions.lean`, state the results of Section 4. A description $k = (h, M)$ is stored in reduced coordinates, and `realUnitY` evaluates $U_k$ on $\mathbb{R}^N$.

```
abbrev FaceKey (r : ℕ) :=
  MaxNN.Caterpillar.Corrections.RealKey (r + 1)

noncomputable def caterpillarDescriptions (r : ℕ) :
    Finset (FaceKey r) :=
  MaxNN.Caterpillar.Corrections.normalizedKeys r

noncomputable def caterpillarFunction {r : ℕ}
    (k : FaceKey r) : Point (r + 1 + 1) → ℝ :=
  MaxNN.Caterpillar.Corrections.realUnitY k

noncomputable def auxiliaryFunction (N : ℕ)
    (j : Fin (MaxNN.Width.W N)) : Point N → ℝ :=
  fun x => MaxNN.Width.feat N x j

def IsCaterpillarExpansion (r : ℕ)
    (K : Finset (FaceKey r)) : Prop :=
  K ⊆ caterpillarDescriptions r ∧
    ∃ (d : FaceKey r → ℝ)
      (a : Fin (MaxNN.Width.W (r + 1 + 1)) → ℝ),
      ∀ x, Maximum (r + 1 + 1) x =
        ∑ k in K, d k * caterpillarFunction k x +
          ∑ j, a j * auxiliaryFunction (r + 1 + 1) j x

noncomputable def auxiliarySpan (N : ℕ) :
    Submodule ℝ (Point N → ℝ) :=
```

```
  Submodule.span ℝ (Set.range (auxiliaryFunction N))

noncomputable def caterpillarSpan (r : ℕ) :
    Submodule ℝ (Point (r + 1 + 1) → ℝ) :=
  Submodule.span ℝ
    ((caterpillarFunction '' ↑(caterpillarDescriptions r))
      ∪ Set.range (auxiliaryFunction (r + 1 + 1)))

noncomputable def caterpillarDim (r : ℕ) : ℕ :=
  FiniteDimensional.finrank ℝ (↑(caterpillarSpan r) /
    (auxiliarySpan (r + 1 + 1)).comap
      (caterpillarSpan r).subtype)
```

### E.3 The main theorems

The assumption `[NeZero N]` expresses positive arity. For $N \geq 3$ the first statement is Theorem 1; the natural-number width definitions also cover $N = 1, 2$. With $N = r + 2$, the next three are Theorem 8, and the last is Theorem 13. All are in `Theorems.lean`.

```
theorem caterpillar_net (N : ℕ) [NeZero N] :
    ∃ net : Network N (firstWidth N) (secondWidth N),
      (∀ x, net.eval x = Maximum N x) ∧
        net.HasRationalWeights ∧ net.HasZeroBiases

theorem caterpillar_expansion_choose_le (r : ℕ)
    {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K)
    {k : ℕ} (hk : 3 ≤ k) (hkN : k ≤ r + 1 + 1) :
    (r + 1 + 1).choose k ≤ 6 ^ k * K.card

theorem caterpillar_expansion_exponential (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    7 ^ (r + 1 + 1) ≤
      (r + 1 + 1 + 1) * 6 ^ (r + 1 + 1) * K.card

theorem caterpillar_expansion_quadratic (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    (r + 1 + 1) * (r + 1) ≤ 6 * K.card

theorem caterpillarDim_ge (r : ℕ) (hr : 1 ≤ r) :
    r.factorial ≤ caterpillarDim r
```

### E.4 The building blocks of Brunck et al.

The definitions of Definition 17, in the indexing of [BHR+26, Section 7.1], are in `Competing/Defs.lean`. An ordering is a bijection from the $N - 1$ positions onto the labels other than $v$, and `Ordering.pos` returns

$\pi_i$ at position $i$, $1 \le i \le N-1$.

```
noncomputable def supportFunction {n : ℕ}
    (P : Set (Fin n → ℝ)) (y : Fin n → ℝ) : ℝ :=
  sSup ((fun p => ∑ k, p k * y k) '' P)

def pyramid {n : ℕ} (q : Fin n → ℝ)
    (P : Set (Fin n → ℝ)) : Set (Fin n → ℝ) :=
  convexHull ℝ (insert q P)

def zonotope {n : ℕ} (a : Fin n → ℝ) (I : Finset ℕ)
    (g : ℕ → Fin n → ℝ) : Set (Fin n → ℝ) :=
  {p | ∃ t : ℕ → ℝ, (∀ i ∈ I, 0 ≤ t i ∧ t i ≤ 1) ∧
    p = a + ∑ i in I, t i • g i}

abbrev BHRSY.Ordering (n : ℕ) (v : Fin n) :=
  Fin (n - 1) ≃ {k : Fin n // k ≠ v}

def BHRSY.Ordering.pos {v : Fin n} (π : Ordering n v)
    (i : ℕ) : Fin n :=
  if h : i - 1 < n - 1 then (π ⟨i - 1, h⟩).1 else v

def BHRSY.hyperplane (v : Fin n) : Set (Fin n → ℝ) :=
  {p | p v = 0 ∧ ∑ k, p k = 1}

noncomputable def BHRSY.prefixSum {v : Fin n}
    (π : Ordering n v) (p : Fin n → ℝ) (i : ℕ) : ℝ :=
  ∑ j in Finset.Ico 1 i, p (π.pos j)

def BHRSY.parallelepiped (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  {p | p ∈ hyperplane v ∧
    (∀ i ∈ M, 0 ≤ prefixSum π p i ∧ prefixSum π p i ≤ 1) ∧
    ∀ i ∈ Finset.Icc 2 (n - 1), i ∉ M →
      -1 ≤ p (π.pos i) ∧ p (π.pos i) ≤ 0}

def BHRSY.paramid (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  pyramid (Pi.single v 1) (parallelepiped v π M)

noncomputable def BHRSY.paramidSupport (v : Fin n)
    (π : Ordering n v) (M : Finset ℕ) : (Fin n → ℝ) → ℝ :=
  supportFunction (paramid v π M)

def BHRSY.lastMarked (M : Finset ℕ) : ℕ :=
  (insert 1 M).max' (Finset.insert_nonempty _ _)

def BHRSY.markedParent (M : Finset ℕ) (i : ℕ) : ℕ :=
```

```
  (insert 1 (M.filter fun j => j < i)).max'
    (Finset.insert_nonempty _ _)
```

The dictionary uses the word $\pi' = (v, \pi_1, \ldots, \pi_{N-1})$ (`Ordering.word`), the cut set $S = M - 1$ (`cutSet`), and the description $(h_{\pi',S}, L_{\pi',S})$ of a word (`caterpillarWordDescription`). The rooted star is `BHRSY.rootedStar`. With $N = r + 2$, `Theorems.lean` states Theorem 18 as follows.

```
theorem bhrsy_parallelepiped_eq_zonotope {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    BHRSY.parallelepiped v π M =
      zonotope (Pi.single (π.pos (BHRSY.lastMarked M)) 1)
        (Finset.Icc 2 (r + 1))
        (fun i => Pi.single (π.pos (BHRSY.markedParent M i)) 1
          - Pi.single (π.pos i) 1)

theorem bhrsy_support_eq {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π M y =
      y v + MaxCaterpillar.relu
        (y (π.pos (BHRSY.lastMarked M)) - y v +
          ∑ i in Finset.Icc 2 (r + 1), MaxCaterpillar.relu
            (y (π.pos (BHRSY.markedParent M i)) - y (π.pos i)))

theorem bhrsy_support_eq_word {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    (BHRSY.cutSet (r := r) M).card = M.card ∧
      ∀ y : Point (r + 1 + 1), BHRSY.paramidSupport v π M y =
        y v + caterpillarFunction
          (caterpillarWordDescription π.word (BHRSY.cutSet M)) y

theorem bhrsy_support_mem_caterpillar {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1))
    (hne : M.Nonempty) :
    ∃ k ∈ caterpillarDescriptions r, ∀ y,
      BHRSY.paramidSupport v π M y = y v + caterpillarFunction k y

theorem bhrsy_support_empty {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π ∅ y =
      y v + BHRSY.rootedStar v (π.pos 1) y

theorem rootedStar_ne_caterpillarFunction {r : ℕ}
    {a c : Fin (r + 1 + 1)} (hac : a ≠ c) {k : FaceKey r}
    (hk : k ∈ caterpillarDescriptions r) :
    BHRSY.rootedStar a c ≠ caterpillarFunction k
```

The same file contains the converse `bhrsy_word_mem`, the special cases `bhrsy_parallelepiped_eq_zonotope_of_nonempty` and `bhrsy_parallelepiped_empty`, the statement `bhrsy_support_empty_not_caterpillar`, and the dichotomy `bhrsy_family`.

### E.5 Proof routes in the formalization

The public theorem `caterpillar_net` uses the rational projection argument directly. The essential normalized descriptions and auxiliary functions span the maximum; applying a $\mathbb{Q}$-linear projection gives rational coefficients on the same support, and the compilation theorem produces the network with rational weights and zero biases. The separate compression theorem first chooses a basis modulo the auxiliary span, giving the second width $q_N + 2$.

The count argument factors over stars of a reduced forest, and the vocabulary-dimension argument separates path functions by a third mixed difference. These are the arguments presented in Appendices C.1 to C.2.

The keyword `noncomputable` allows classical definitions of real quantities. The theorem is an exact representation statement; it does not by itself assert an efficient algorithm for extracting numerical weights.

### E.6 Verification scope

The repository's historical verification certificate, dated 24 October 2026, records a build with Lean 4.4.0 and checks 122 public statements against their exact types. Its permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. The checked interface includes dictionary sufficiency, rational expansions, network correctness with rational weights and zero biases, the width bounds, the compilation and compression theorems, the Wang–Sun and Koutschan–Moser–Ponomarchuk–Schicho representations, and the network theorems for sums of maxima and CPWL functions with their rationality and bias statements. The equation and theorem links identify supporting declarations for the mathematical arguments in the paper. The certificate and its source hashes are available in the linked verification record. The subsequent consolidation of the Lean modules is not covered by that certificate: the current sources require an updated exact-statement manifest and a fresh verification run. In particular, the common local test and the source-orientation, matching-number, and arbitrary-star extensions are mathematical arguments in this paper, not claims checked by the historical certificate. The mathematical arguments are provided in the main text and Appendices A to C; the formalization is an additional certificate of the formal statements.

## AI methodology statement

Claude Code and Codex were used for mathematical exploration and proof search, development of the Lean formalization, and drafting and revision of the manuscript. Their use materially affected the construction and proofs in Sections 3 to 8 and Appendices A to E. The author set the research objectives, directed the searches, and revised the mathematical presentation. The accompanying Lean certificate checks the formal statements and their axiom dependencies; it does not certify the prose exposition, literature comparison, or claims of originality. The author is responsible for the contents of the paper.

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